ECO612: Applied Macroeconomics — Group Assignment

An Empirical Assessment of Monetary and Fiscal Policy in India, 2014–2026

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Data, definitions and conventions

All series come from official sources, and the full data log is in Appendix A. The conventions below are used in every question.

Real GDP and consumption. We use quarterly real GDP, nominal GDP and real private final consumption expenditure (PFCE) from the IMF's *International Financial Statistics* (FRED series `NGDPRNSAXDCINQ`, `NGDPNSAXDCINQ`, `NCPRNSAXDCINQ`), covering 2004Q2–2026Q2. From 2022Q2 the series is MoSPI's new base (2022–23 prices); earlier quarters are an IMF-linked back series. The IMF link implies FY2022–23 growth of about 9.6%, against the official 7.6%. We therefore re-link the old segment with a ratio splice. The RBI *Handbook* (Tables 1–2) publishes 2022–23 GDP under both bases (Rs 161.6 vs Rs 261.2 lakh crore at constant prices), and we rescale the old segment so that FY2021–22 equals the old-base value times this ratio. By construction the splice gives 2022–23 growth equal to the official old-base rate (7.6%) when annual totals are used. Summing the IMF's new-base quarters for 2022–23 gives a total 0.23% above the *Handbook* annual figure, so annual growth computed from the spliced quarterly series is 7.85% in 2022–23, slightly above the official 7.6%. Other years are close to the official prints: −5.8% in 2020–21, 9.7% in 2021–22 and 7.8% in 2025–26. We keep the quarterly data unaltered rather than force the quarters to the annual total; the 0.25 pp discrepancy affects only the level of 2022–23 growth, not the cyclical analysis.

Seasonal adjustment. The data are not seasonally adjusted. For the old-base segment (72 quarters) we use robust STL (period 4), which allows slowly evolving seasonality and down-weights pandemic outliers. For the new-base segment, which is too short for STL, we estimate fixed quarter effects by OLS with a trend. The variable y_t denotes $100 \times$ log of seasonally adjusted (SA) real GDP. Growth g_t is the year-on-year change in y_t.

Inflation and the policy rate.

- π_t is year-on-year CPI-Combined inflation, the RBI's target variable, as a monthly series or a quarterly average.
- The CPI base changes from 2012=100 to 2024=100 in January 2026. We use the published year-on-year rates on each base.
- i_t is the policy repo rate, measured as a daily step function averaged over the quarter (RBI *Handbook* Table 40, plus the 7 October 2026 hike to 5.50%).

Status of the data. 2024–25 national accounts are first revised estimates, 2025–26 are provisional estimates, and the 2025–26 quarters and 2026Q2 are early estimates. Budget figures are actuals up to 2024–25, revised estimates (RE) for 2025–26 and budget estimates (BE) for 2026–27. Nothing in the analysis is a forecast except the BE year, which is labelled.

Inference. Unless stated otherwise, standard errors are Newey–West HAC (lag 4 for quarterly regressions, lag 1 for annual ones). The one exception is the annual distributed-lag model in Q3(b), which reports OLS standard errors (explained there). Significance is marked * 10%, ** 5%, *** 1%. Every regression table reports a serial-correlation test. Augmented Dickey–Fuller (ADF) tests for all regression variables are collected in Table 15 (Appendix A). All code is in Appendix B.

1 Q1. India's macroeconomic position

(a) Growth, inflation and the repo rate

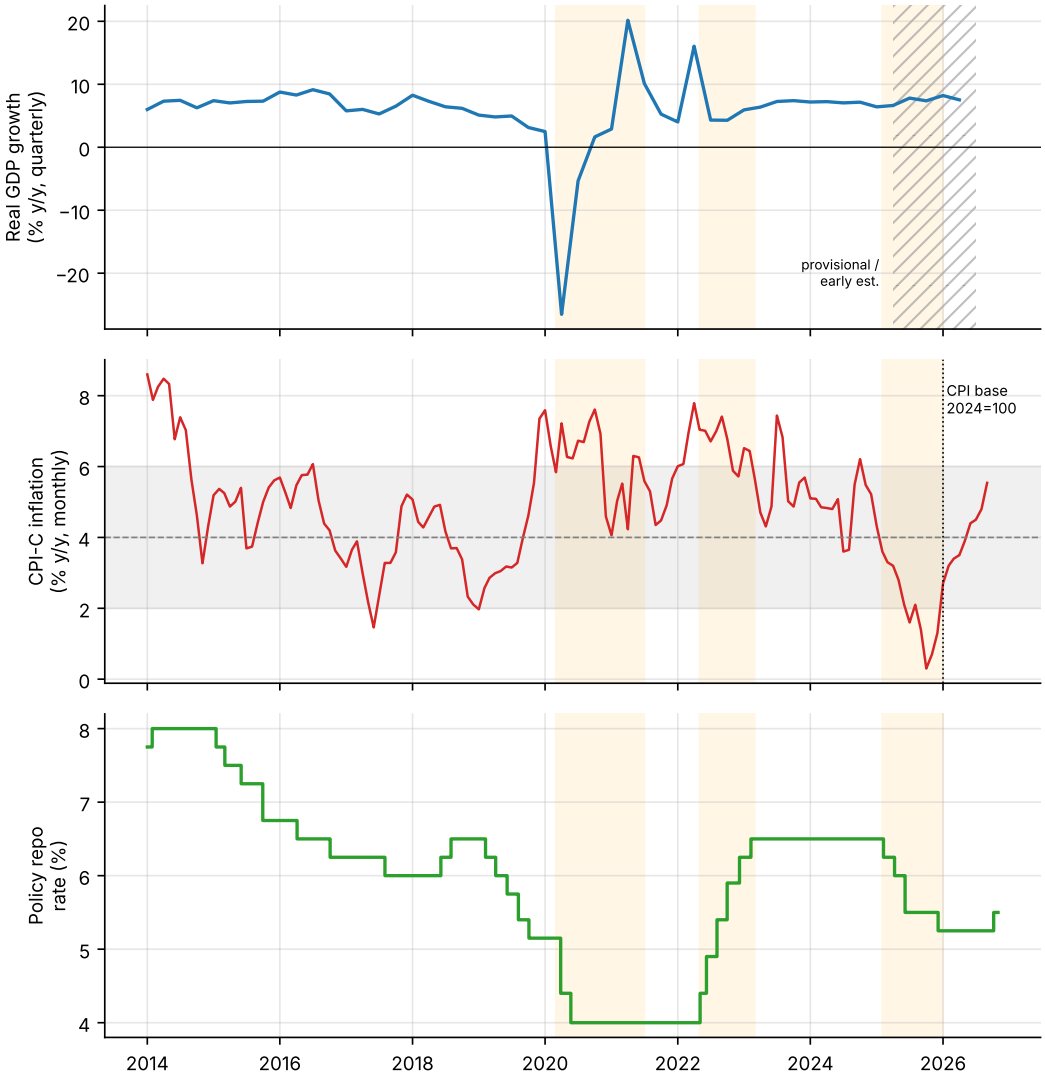


Figure 1: Real GDP growth (SA, % y/y), CPI-Combined inflation (monthly, % y/y, with the 2–6% tolerance band) and the policy repo rate, 2014–2026. Shaded areas mark the pandemic, the 2022–23 tightening and the 2025 easing. Hatching in the top panel marks provisional and early national-accounts estimates (2025–26 quarters and 2026Q2).

Table 1: Phase averages (%). Growth: SA real GDP, y/y. Repo: quarter average.

Phase (mean, %)	Growth	CPI infl.	Repo (avg)
Pre-pandemic (2018Q2–2020Q1)	5.05	4.10	5.89
Pandemic (2020Q2–2021Q1)	-6.83	6.18	4.06
Recovery (2021Q2–2022Q1)	9.89	5.51	4.00
Tightening (2022Q2–2023Q1)	7.65	6.66	5.50
Hold (2023Q2–2024Q4)	7.09	5.18	6.50
2025 easing (2025Q1–2025Q4)	7.06	2.23	5.79
2026 (2026Q1–2026Q3)	7.87	3.99	5.25

We identify three turning points.

(1) The pandemic collapse, 2020Q2. Growth had already slowed from about 8% in 2018 to 3.1% in

2019Q4. That pre-pandemic slowdown was the main reason the RBI cut the repo rate by 135 bp during 2019. In 2020Q2 output fell 26.5% year on year, the deepest contraction in the series. The 2021Q2 rebound of +20.2% is mostly a base effect.

The RBI cut the repo rate from 5.15% to 4.00% (115 bp) between March and May 2020, taking cumulative easing since February 2019 to 250 bp. It then held the rate at 4% for two years.

Inflation did not fall with output; it averaged 6.6% in 2020Q2 and 6.2% across the pandemic year. The shock therefore hit both supply and demand, and it left policymakers with a sharply negative output gap alongside inflation above target.

(2) The inflation peak and tightening, 2022. CPI inflation peaked at 7.79% in April 2022 and averaged 6.8% over April–December 2022. This reflected the global commodity shock and the after-effects of supply disruption. The RBI raised the repo rate by 250 bp between May 2022 and February 2023 (4.00% to 6.50%), after an off-cycle start, and real growth held at 7.6% in 2022–23.

The significance of this episode is that tightening happened while the output gap was still negative under most measures (Q1(b)). It was a response to inflation persistence, not to overheating.

(3) Disinflation and easing, 2025, followed by a reversal in 2026. Inflation fell from 5.2% in December 2024 to 0.3% in October 2025 and averaged 2.2% across 2025. Two forces drove it. The main one was food deflation. The second was tax policy: the GST rate rationalisation ("GST 2.0"), effective 22 September 2025, replaced the 12% and 28% slabs with two main rates of 5% and 18% and cut rates on many everyday goods. That produced a one-off fall in the *price level*, which depresses year-on-year inflation for twelve months and then drops out. That is below the 4% target and, for part of the year, below the 2% lower band. The RBI cut the repo rate by 125 bp (6.50% to 5.25%) between February and December 2025, including a 50 bp cut in June, while growth stayed near 7.1%.

The turning point is incomplete. Inflation rebounded to 5.54% by September 2026 (new CPI base), and the RBI raised the repo rate to 5.50% on 7 October 2026. The 2025 easing therefore responded to a large but transitory inflation undershoot, partly caused by food prices and partly by the tax cut, rather than to weak demand. The base effect from the GST cut is one reason inflation rose through 2026.

(b) Potential output and the output gap

We estimate the gap $\tilde{y}_t = y_t - y_t^*$ (% of potential) on 2004Q2–2026Q2 using three methods.

1. **HP filter** ($\lambda = 1600$). The standard filter treats the 2020Q2 collapse as partly permanent and pulls the trend down before 2020. This produces an implausibly large pre-pandemic gap of +3.2% in 2019Q4. Our baseline, denoted "HP" from here on, applies the filter to a series in which 2020Q2–2021Q2 are linearly interpolated, then measures the gap against actual output. We report the standard HP filter alongside it.

2. **Hamilton (2018) regression filter.** We regress y_{t+8} on $y_t, \ldots, y_{t-3}$; the gap is the two-year-ahead forecast error. As a robustness check we re-estimate the regression excluding every observation whose dependent variable or regressors fall in 2020Q2–2021Q2, and apply those coefficients to the full sample ("Hamilton (ex-COVID est.)").

3. **Deterministic linear trend with a level break in 2020Q2.** The fitted trend growth is 7.1% a year, and the estimated permanent level loss is 14.8 log points. We also report a linear trend without a break.

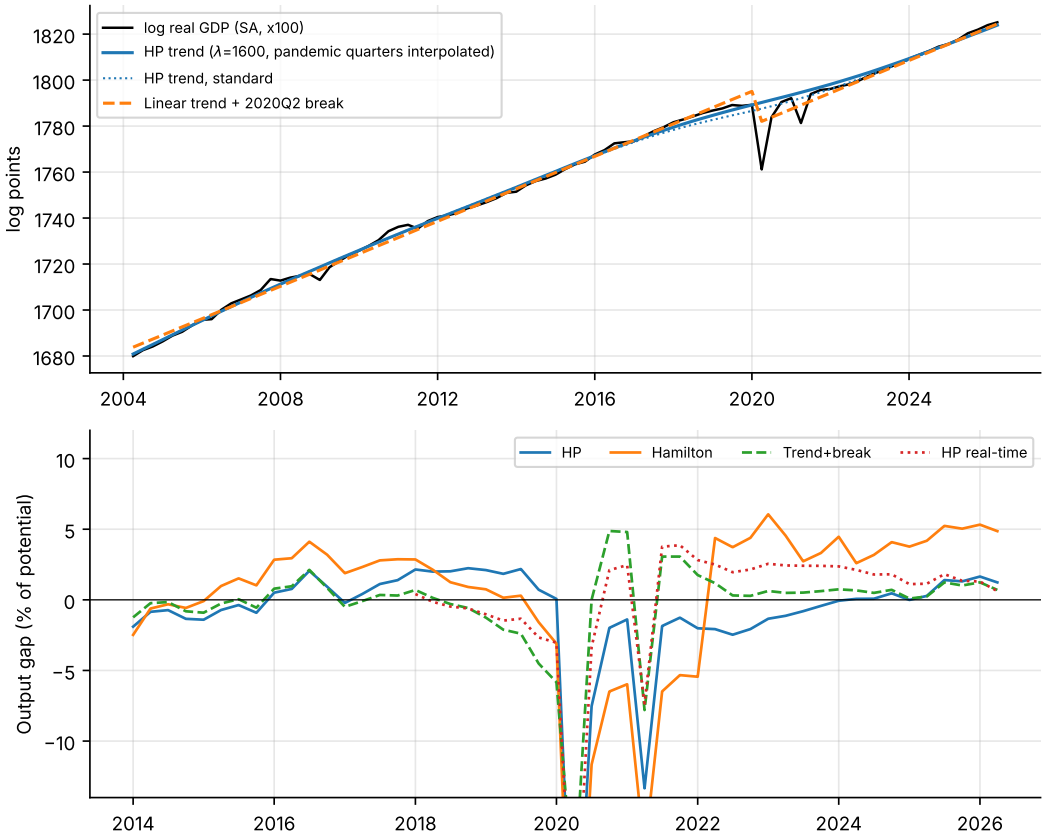


Figure 2: Top: log SA real GDP with the HP and broken-trend estimates of potential. Bottom: output gap estimates, 2014–2026 (the y-axis is truncated at −14; the 2020Q2 trough is about −21 to −33%).

Table 2: Output gap by episode (% of potential) and sensitivity to method, smoothing, sample and end-point.

Episode	HP	HP (standard)	Hamilton	Hamilton (ex-COVID est.)	Trend+break	Linear trend	HP (λ=400)	HP (from 2011)	HP (end 2025Q4)	HP real-time
2019Q4 (pre-COVID)	0.72	3.20	-1.58	-2.76	-4.51	3.28	0.31	0.71	0.67	-2.66
2020Q2 (trough)	-29.13	-26.44	-32.86	-34.01	-20.91	-27.52	-29.17	-29.14	-29.18	-26.09
FY2021-22 avg	-4.62	-2.31	-8.79	-10.00	0.02	-5.49	-3.79	-4.62	-4.66	0.78
FY2022-23 avg	-1.99	-0.41	4.64	3.28	0.60	-4.10	-1.02	-1.99	-1.98	2.29
2025 avg	0.74	0.57	4.56	2.80	0.64	-1.84	0.24	0.75	1.29	1.35
2026Q2 (latest)	1.23	0.62	4.86	3.06	0.70	-1.07	0.10	1.24	–	0.62

Notes: HP = baseline (pandemic quarters interpolated before filtering). "Hamilton (ex-COVID est.)" uses Hamilton-regression coefficients estimated without the pandemic quarters. "HP real-time" is the one-sided filter that uses only data available up to each quarter. "HP (end 2025Q4)" re-estimates the baseline without the 2026 observations.

Comparison across episodes.

- All methods agree on the sign at the extremes: a gap of about −21 to −33% in 2020Q2 and negative slack in 2021–22, ranging from 0 (broken trend) to −8.8% (Hamilton).
- They disagree most about the post-pandemic level. For 2025 the gap is +0.7% (HP), +0.6% (broken trend), +4.6% (Hamilton) and −1.8% (linear trend without a break). The main correlations are 0.88 between HP and Hamilton and 0.78 between HP and the broken trend.
- The Hamilton gap is, in effect, two-year growth relative to the growth the regression predicts. For 2025 the forecasts use 2023 data, which are already post-shock, so the large positive gap is *not* a forecast from pre-pandemic levels. It has two sources. First, 2023–26 growth (about 7.3–7.8% a year) exceeds the sample-average growth the regression embeds, which the 2019 slowdown and the pandemic pull down. Second, the pandemic quarters distort the estimated coefficients. Removing them from the estimation lowers the 2025 gap from +4.6% to +2.8% (Table 2), so about two-fifths of the original

Hamilton gap reflects the pandemic observations. The remainder is genuine: on Hamilton's measure, recent growth is faster than India's average.

- The unbroken linear trend assumes the pandemic loss will be fully recovered, so it implies persistent slack.
- The broken trend and the HP filter both effectively accept the 2020 level loss as permanent. Whether the economy was overheating in 2025 is therefore mainly a question of how much of the pandemic loss was permanent.

Sensitivity.

- *Smoothing parameter.* With $\lambda = 400$ the 2025 gap shrinks to $+0.2\%$.
- *Sample start.* Starting the sample in 2011 changes nothing.
- *End-point.* When the 2026 observations are dropped, the 2025 HP gap rises from $+0.74\%$ to $+1.29\%$. Adding four quarters of data revises the estimate by 0.55 pp.
- *Real time.* The one-sided filter put the 2022–23 gap at $+2.3\%$, whereas the two-sided estimate is -2.0%, a revision of more than 4 pp that flips the sign. Measured in real time, the economy looked overheated during the tightening; with hindsight it was still recovering.

Both baseline gap measures are stationary: ADF p-values are < 0.001 (HP) and 0.002 (Hamilton). The ex-COVID Hamilton gap does not reject a unit root ($p = 0.23$; Table 15), so it is a robustness check rather than a replacement.

Implication. Slack is measured with an uncertainty band of roughly ± 3 pp around a central estimate close to zero for 2024–26. Conclusions that depend on the sign of the 2025 gap are therefore fragile. Rules and reaction functions in Q2 should be checked across gap measures, and real-time policy judgements should be read with end-point revisions in mind.

(c) Inflation equation

We estimate $\pi_t = \alpha + \rho\pi_{t-1} + \beta_0\tilde{y}_t + \beta_1\tilde{y}_{t-1} + u_t$ on quarterly data for 2014Q2–2026Q2 ($T = 49$), where π_t is the quarterly average of year-on-year CPI-C inflation in percent.

Table 3: Inflation equations. Dependent variable: π_t. HAC (4) standard errors in parentheses.

	(1) HP, no lag	(2) HP	(3) Hamilton	(4) Trend+brk	(5) HP ex-COVID
Constant	1.424^{***}	1.525^{***}	1.506^{***}	1.288^{***}	2.023^{***}
	(0.389)	(0.401)	(0.381)	(0.397)	(0.512)
π_{t-1}	0.681^{***}	0.656^{***}	0.674^{***}	0.713^{***}	0.569^{***}
	(0.070)	(0.074)	(0.071)	(0.073)	(0.091)
$\tilde{y}_t$	-0.043^{**}	-0.039^{**}			-0.340^{**}
	(0.019)	(0.016)			(0.132)
$\tilde{y}_{t-1}$		-0.027^{**}			0.058
		(0.011)			(0.048)
$\tilde{y}_t$ (Hamilton)			-0.019^{*}		
			(0.010)		
$\tilde{y}_{t-1}$ (Hamilton)			-0.023^{**}		
			(0.011)		
$\tilde{y}_t$ (trend)				-0.043^{***}	
				(0.016)	
$\tilde{y}_{t-1}$ (trend)				-0.031	
				(0.026)	
Observations	49	49	49	49	44
R^2 (adj.)	0.576	0.573	0.574	0.570	0.564
AIC	143.3	144.6	144.4	144.9	131.2
Joint F: gap terms = 0 (p)	–	0.010	0.003	0.010	0.026
$\beta_0 + \beta_1$	-0.043	-0.066	-0.042	-0.074	-0.282
Breusch–Godfrey(4) p	0.027	0.022	0.023	0.063	0.094

Results. Inflation is persistent ($\hat{\rho} = 0.66$, s.e. 0.07). Holding slack constant, a 1 pp shock to inflation decays with a half-life of about 1.6 quarters.

Contrary to a textbook Phillips curve, both gap coefficients are *negative*: $\hat{\beta}_0 = -0.039$ (0.016) and $\hat{\beta}_1 = -0.027$ (0.011).

- *Joint significance.* The gap terms are jointly significant (F test, $p = 0.010$).
- *Combined association.* $\hat{\beta}_0 + \hat{\beta}_1 = -0.066$ (s.e. 0.021, 95% CI $[-0.107, -0.025]$). A 1 pp larger gap sustained over two quarters goes with 0.07 pp *lower* inflation.
- *Long-run association.* $(\beta_0 + \beta_1)/(1-\rho) = -0.19$ (0.06).
- *Magnitude.* Even at its estimated value the association is economically small: a gap of $+1\%$ moves inflation by less than 0.2 pp in the long run.
- *Robustness.* The result holds with the Hamilton gap (combined -0.042) and the broken-trend gap (-0.074). Excluding 2020Q2–2021Q2 makes it larger (-0.28).

Does lagged slack add explanatory power? The coefficient on $\tilde{y}_{t-1}$ is individually significant (HAC $p = 0.017$). However, adding it lowers adjusted R^2 (0.576 to 0.573) and raises the AIC (143.3 to 144.6), so its marginal predictive contribution is negligible.

Diagnostics. A Breusch–Godfrey test rejects no serial correlation at lag 4 ($p = 0.02$), as expected with overlapping year-on-year data. This is why we use HAC inference. Inflation itself does not reject a unit root (ADF $p = 0.32$). With 49 observations this test has low power, and the inflation-targeting regime makes a mean-reverting process more plausible.

Why this is not a causal Phillips curve.

1. *Supply shocks dominate CPI-C.* Food has a weight of about 46% in the 2012 basket. The pandemic, the 2022 commodity shock and the 2024–25 vegetable cycle moved inflation and output in opposite directions, creating a negative correlation that says nothing about the demand-side slope. The sample combines a negative-gap/high-inflation episode (2022) with a positive-gap/very-low-inflation episode (2025).

2. *Endogenous policy.* Monetary policy responds to expected inflation (Q2) and offsets demand pressure. This flattens or reverses the reduced-form slope even if the structural slope is positive.

3. *Measurement error.* The gap is mismeasured, as shown in Q1(b), which biases coefficients toward zero.

4. *Omitted expectations.* The equation omits inflation expectations, the exchange rate and supply-shock controls such as food and fuel inflation.

Identifying the structural slope would require at least supply-shock controls, core inflation as the dependent variable, or instruments for the gap. The robust conclusion is a negative one: headline inflation in India since 2014 has been driven far more by supply shocks and its own persistence than by domestic slack.

2 Q2. Did monetary policy respond systematically?

(a) A Taylor-type benchmark

The rule is $i_t^{TR} = r^* + \pi_t + \theta_\pi(\pi_t - \pi^*) + \theta_y \tilde{y}_t$, with $\pi^* = 4\%$ (the statutory CPI target). We use three calibrations:

- **Rule A (baseline):** $r^* = 1.5\%$, $\theta_\pi = \theta_y = 0.5$, HP gap.
  - $r^* = 1.5\%$ lies inside the 1.4–1.9% range that RBI staff estimate for India's natural real rate in 2023–24 Q4 (RBI *Bulletin*, July 2024 [6]), up from 0.8–1.0% in 2021–22.
  - $\theta_\pi = \theta_y = 0.5$ are Taylor's (1993) original weights, which imply a long-run nominal response of 1.5 to inflation.
- **Rule B (hawkish):** $r^* = 1.0\%$, $\theta_\pi = 1.0$, $\theta_y = 0.5$, Hamilton gap. This is a strong inflation response combined with a gap measure that sees more overheating.
- **Rule C (higher neutral rate, more weight on output):** $r^* = 1.9\%$, $\theta_\pi = 0.5$, $\theta_y = 1.0$, broken-trend gap.

Rules B and C each change several elements at once, which shows the plausible range but not which element drives it. Table 5 therefore also changes one element of Rule A at a time.

Inflation is measured as backward-looking year-on-year headline CPI. For 2026Q3, GDP is not yet released, so the 2026Q2 gap is carried forward.

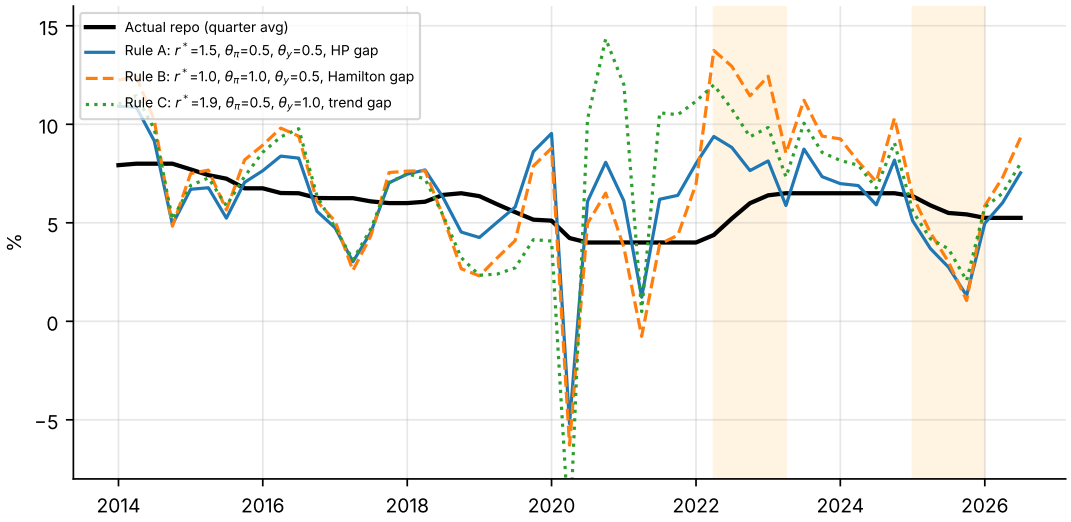


Figure 3: Actual repo rate (quarter average) and rule-implied rates, 2014Q1–2026Q3. Shaded: 2022–23 tightening and 2025 easing.

Table 4: Actual vs rule-implied policy rates by episode (means, %). $i - i^X$ is the policy gap; a negative value means policy was looser than the rule.

Episode (means, %)	Repo	π	Gap(HP)	i^A	$i - i^A$	i^B	$i - i^B$	i^C	$i - i^C$
Pre-tightening 2021Q4–2022Q1	4.00	5.68	-1.64	7.20	-3.20	5.67	-1.67	10.83	-6.83
Tightening 2022Q2–2023Q1	5.50	6.66	-1.99	8.50	-3.00	12.64	-7.15	10.49	-5.00
Hold 2023Q2–2024Q4	6.50	5.18	-0.26	7.13	-0.63	9.13	-2.63	8.27	-1.77
Easing 2025Q1–2025Q4	5.79	2.23	0.74	3.21	2.58	3.73	2.06	3.88	1.91
2026Q1–2026Q2	5.25	3.52	1.44	5.50	-0.25	6.58	-1.33	6.14	-0.89
2026Q3†	5.25	4.95	1.23	7.54	-2.29	9.32	-4.07	8.02	-2.77

† 2026Q3: inflation for July–September 2026; the gap is the 2026Q2 value carried forward. The repo average excludes the 7 October 2026 hike.

Table 5: One-at-a-time sensitivity: Rule A with a single element changed (episode means, %).

Change from Rule A	i^{TR} tight.	$i - i^{TR}$ tight.	i^{TR} easing	$i - i^{TR}$ easing
Rule A (baseline)	8.50	-3.00	3.21	2.58
$r^* = 1.0$	8.00	-2.50	2.71	3.08
$r^* = 1.9$	8.90	-3.40	3.61	2.18
$\theta_\pi = 1.0$	9.83	-4.33	2.32	3.47
$\theta_y = 1.0$	7.51	-2.01	3.58	2.21
Hamilton gap	11.81	-6.32	5.12	0.67
Trend+break gap	9.79	-4.30	3.16	2.63

2022–23 tightening. Policy was well behind every rule. When tightening began (2021Q4–2022Q1) Rule A prescribed 7.2% against an actual 4.0%. Over the tightening year the repo averaged 5.5% against 8.5% (A), 12.6% (B) and 10.5% (C), giving policy gaps of −3.0, −7.2 and −5.0 pp.

The gap closed mainly because inflation fell, not because the repo rose. By the 2023–24 hold the shortfall was down to −0.6 pp (A). All three rules therefore agree that tightening came late. Its *direction* was right, and its size (250 bp) was about the size of the gap Rule A identified at the start.

2025 easing. The pattern reverses. With inflation averaging 2.2% and a small positive gap, Rule A prescribed 3.2% while the repo averaged 5.8%, a policy gap of +2.6 pp. Rules B (+2.1) and C (+1.9) agree. On a backward-looking basis the RBI eased *less* than a mechanical rule would have. That is consistent with

looking through a disinflation it expected to reverse: food prices, and from September 2025 a one-off GST cut that lowered the price level without signalling weaker demand.

The 125 bp of cuts closed about 38% of the gap between the pre-easing repo (6.50%) and the 2025 Rule-A rate (3.21%). The ex-post evidence supports the caution: by 2026Q3, with inflation at 5.0% (quarter average) and a gap of +1.2%, Rule A prescribes 7.5% and Rule B 9.3%, against an actual 5.25%. The October 2026 hike to 5.50% moves in the direction all three rules indicate.

Caveats.

1. The RBI sets policy on *projected* inflation two to four quarters ahead, so a backward-looking rule overstates the required response to transitory food shocks.
2. The rule level depends one-for-one on r^*: a range of 1.0–1.9% shifts i^{TR} by 0.9 pp.
3. The gap measure matters most. Table 5 shows that switching Rule A to the Hamilton gap alone raises the tightening-year prescription by 3.3 pp and the 2025 prescription by 1.9 pp, cutting the 2025 policy gap from +2.6 to +0.7 pp. No other single change moves the 2025 policy gap by more than 0.9 pp. Rule B is 4.1 pp above Rule A in the tightening year and 2.0 pp above in the hold, but only 0.5 pp above in 2025 and 1.3 pp in 2026, because its stronger inflation response ($\theta_\pi = 1$) pushes it *down* when inflation is below target.

Across all single changes the sign of the policy gap never changes: policy was tighter than the rule in 2025 and looser in 2022–23 (Table 5).

(b) Estimated reaction function

We estimate $i_t = \alpha + \rho i_{t-1} + \beta_\pi \pi_t + \beta_y \tilde{y}_t + u_t$ on quarterly data for 2014Q3–2026Q2 ($T = 48$).

Table 6: Monetary policy reaction functions. Dependent variable: repo rate (quarter average). HAC(4) s.e. Long-run responses $\beta/(1-\rho)$ with delta-method s.e.

	(1) Baseline	(2) Hamilton	(3) +lags	(4) lags only	(5) ex-COVID
Constant	0.076	0.250	0.090	0.006	0.281
	(0.160)	(0.205)	(0.171)	(0.186)	(0.288)
i_{t-1}	0.921***	0.886***	0.888***	0.926***	0.908***
	(0.028)	(0.030)	(0.032)	(0.044)	(0.046)
π_t	0.078**	0.077***	0.002		0.056**
	(0.032)	(0.027)	(0.033)		(0.022)
π_{t-1}			0.114***	0.081**	
			(0.042)	(0.041)	
$\tilde{y}_t$ (HP)	0.031***		0.035***		−0.033
	(0.005)		(0.006)		(0.039)
$\tilde{y}_{t-1}$ (HP)			0.012**	0.013*	
			(0.005)	(0.008)	
$\tilde{y}_t$ (Hamilton)		0.031***			
		(0.006)			
Observations	48	48	48	48	42
R^2 (adj.)	0.950	0.962	0.961	0.942	0.937
AIC	4.8	-8.8	-5.2	11.6	1.8
LR π response	0.98	0.67	1.05	1.10	0.61
(s.e.)	(0.39)	(0.27)	(0.28)	(0.39)	(0.32)
LR gap response	0.39	0.27	0.42	0.18	-0.36
(s.e.)	(0.13)	(0.06)	(0.08)	(0.05)	(0.47)
BIC	12.3	-1.3	6.0	19.0	8.7
Breusch–Godfrey(4) p	0.003	0.028	0.003	0.002	0.005

Baseline (column 1). Interest-rate smoothing dominates: $\hat{\rho} = 0.921$ (0.028). Only 8% of the gap between the current rate and the desired rate is closed each quarter, a half-life of about 8.4 quarters.

The short-run coefficients are $\hat{\beta}_\pi = 0.078$ (0.032) and $\hat{\beta}_y = 0.031$ (0.005), both significant. The implied long-run responses are:

- to inflation: $\beta_\pi/(1-\rho) = 0.98$ (s.e. 0.39, 95% CI $[0.22, 1.74]$);
- to the gap: $\beta_y/(1-\rho) = 0.39$ (0.13).

A permanent 1 pp rise in inflation eventually raises the repo by about 1 pp, and a 1 pp larger gap raises it by about 0.4 pp.

Taylor principle. The point estimate of the long-run inflation response is almost exactly 1. A Wald test of $\beta_\pi + \rho = 1$ (equivalent to a long-run response of one) gives $t = -0.05$, so we cannot reject that real rates are held *constant* in response to inflation. Nor can we confirm that they rise.

The estimate is lower with the Hamilton gap (0.67, s.e. 0.27) and excluding the pandemic (0.61, s.e. 0.32), but in neither case can we reject a long-run response of one ($t = -1.20$ and -1.24; even a one-sided 10% test needs $t < -1.28$). In the lag models (columns 3–4) the long-run response is 1.05 (0.28) and 1.10 (0.39). Across all five specifications, then, the data are consistent with a long-run inflation response of about one. They neither confirm nor reject the Taylor principle.

These magnitudes must be interpreted with care. With ρ close to 1, long-run ratios are imprecise. The

residuals are serially correlated (Breusch–Godfrey $p < 0.01$), so a lagged dependent variable combined with autocorrelated errors may bias $\hat{\rho}$ upward and the long-run responses with it. Since the repo rate does not reject a unit root (ADF $p = 0.14$) while its change does ($p = 0.008$), this is a near-integrated regression. The results describe systematic behaviour; they are not structural parameters.

Was the 2025 easing in line with the estimated rule? We simulate column 1 dynamically from the actual 2024Q4 repo rate, feeding in actual inflation and the HP gap. The simulated repo falls to 5.62% by 2025Q4 and 5.52% by 2026Q2, against actual averages of 5.43% and 5.25%. The RBI therefore cut about 20–30 bp *more* than its own 2014–2026 behaviour predicts, and it did so mainly in mid-2025: the one-step residuals are -0.26 pp in 2025Q2 and -0.17 pp in 2025Q3, the quarters of the 50 bp June cut. So the two benchmarks disagree. Against the Taylor rule, 2025 easing was cautious. Against the RBI's own history, it was somewhat faster than usual, which suggests the MPC gave weight to the forecast of below-target inflation rather than only to past outturns.

(c) Lags, leads and the nature of the response

Adding lags. Column 3 adds π_{t-1} and $\tilde{y}_{t-1}$. The two lagged terms are jointly significant (F test, $p = 0.010$). The model improves on every criterion: adjusted R^2 rises from 0.950 to 0.961 and BIC falls from 12.3 to 6.0.

Once lagged inflation is included, *current* inflation becomes irrelevant (0.002, s.e. 0.033), while lagged inflation is significant (0.114, s.e. 0.042). The gap enters both contemporaneously (0.035^{***}) and with a lag (0.012^{**}). The implied long-run responses barely change (inflation 1.05, s.e. 0.28; gap 0.42, s.e. 0.08), so lags redistribute the timing without changing the overall strength.

The lead–lag correlations agree. The quarterly change in the repo is most correlated with *last quarter's* inflation (0.40, against 0.28 contemporaneous and 0.06 two quarters ahead), and with the *current* gap (0.27). A Granger test of $\Delta\pi \to \Delta i$ ($p = 0.17$) is suggestive but not significant.

Inflation leads policy by about one quarter, and output enters in real time. This is consistent with three explanations:

1. *Delayed response and data lags.* CPI is published about two weeks after the month ends, and the MPC meets every two months, so a quarter's inflation is fully seen only at the next meeting.

2. *Policy inertia and deliberate gradualism.* The 2022 cycle started a quarter or two after inflation breached 6%, and smoothing ($\rho \approx 0.9$) spreads the adjustment over many quarters.

3. *A forward-looking rule proxied by lagged data.* If the RBI reacts to its inflation *projection*, and projections are formed partly from recent outturns, lagged inflation will appear in the regression even if the true rule is forward-looking.

We cannot fully separate these explanations without real-time forecast data.

Reaction function vs causal effect. A reaction function is a regression of the policy instrument on the state of the economy. It describes how policy *responds* to inflation and output, so its coefficients run from the economy to policy. It cannot tell us the effect of policy *on* inflation or output.

Estimating that effect requires identifying exogenous policy shocks, the residual u_t in a correctly specified rule, with methods such as high-frequency surprises around MPC announcements, narrative identification or structural VARs. Simply regressing inflation on the repo rate would confound the two directions of causation. Because policy tightens when inflation is high, such a regression could even show a positive "effect" of rates on inflation, a version of the price puzzle.

3 Q3. Transmission to banks and borrowers

Data. We use weighted average lending rates (WALR) on fresh and outstanding rupee loans, the median one-year marginal cost of funds based lending rate (MCLR), and the weighted average domestic term-deposit rate (WADTDR) on outstanding deposits. All are for scheduled commercial banks, end-March values from RBI *Handbook* Table 59 (FY2013–14 to May 2026). We add monthly values for February 2025–August 2026 from the RBI's "Lending and Deposit Rates of SCBs" releases, which include WADTDR on fresh deposits. The monthly time series before 2025 are distributed only as spreadsheets on RBI's document server, which we could not access. The distributed-lag analysis in part (b) is therefore annual. This is the main data limitation of the report: annual end-March changes cannot resolve within-year timing, and 11 observations give the asymmetry tests little power. Appendix A explains how to extend the analysis to monthly data from RBI's online database, which would be our first improvement.

(a) Episode pass-through

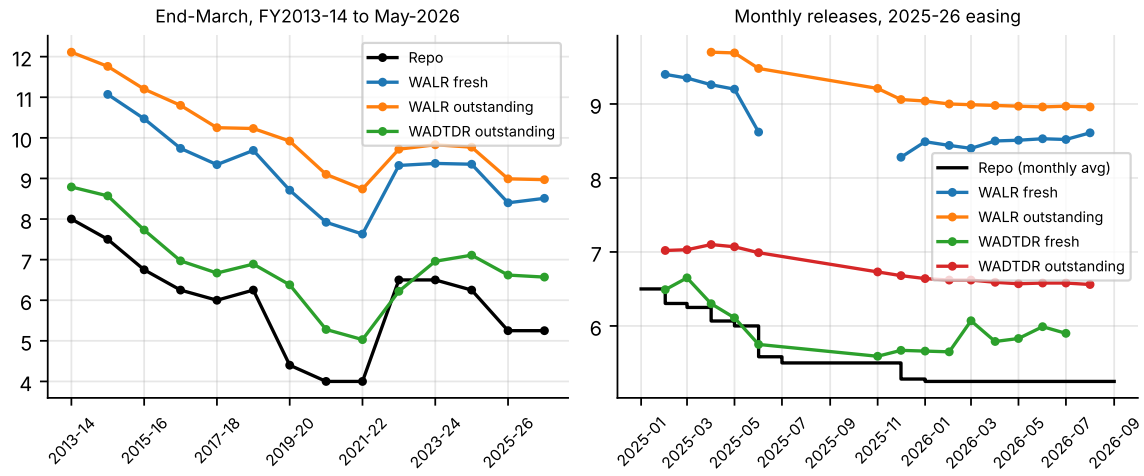


Figure 4: Repo, lending and deposit rates. Left: end-March, 2013–14 to May 2026. Right: monthly releases during the 2025–26 easing (gaps are months not retrieved).

Table 7: Changes (bp) and simple pass-through ratios $\Delta R/\Delta i$ by measurement window.

Window	Δrepo	ΔWALR-F	PT	ΔWALR-O	PT	ΔMCLR	PT	ΔWADTDR-O	PT
T1: Mar-22→Mar-23	250.00	169.00	0.68	98.00	0.39	130.00	0.52	119.00	0.48
T2: Mar-22→Mar-24	250.00	174.00	0.70	109.00	0.44	155.00	0.62	193.00	0.77
E1: Mar-25→Mar-26	-100.00	-95.00	0.95	-78.00	0.78	-60.00	0.60	-49.00	0.49
E2: Mar-24→May-26	-125.00	-86.00	0.69	-86.00	0.69	-15.00	0.12	-39.00	0.31
E3: Feb-25→Aug-26‡	-125.00	-79.00	0.63	-74.00	0.74	–	–	-46.00	0.37

WALR-F/O: WALR on fresh/outstanding loans. MCLR: one-year median. WADTDR-O: deposit rate on outstanding term deposits. ‡ Monthly releases. Δrepo is measured over each series' own window (repo at the start of the first month to the end of the last). The outstanding WALR series starts in April 2025, after the February cut, so its window covers −100 bp, not −125 bp. The fresh WADTDR fell 59 bp between February 2025 and July 2026 (ratio 0.47).

Windows.

- *Tightening* (+250 bp, May 2022–February 2023). T1 measures from end-March 2022 to end-March 2023, roughly contemporaneous with the hikes. T2 extends to end-March 2024, giving about 13 months after the last hike.

- *Easing* (-125 bp, February–December 2025). E1 runs from end-March 2025 to end-March 2026 (after the first cut, so $\Delta i = -100$). E2 runs from March 2024 (before any cut) to May 2026. E3 uses the monthly release data from February 2025 to August 2026.

Results.

- *Fresh loans react fastest.* Pass-through to new-loan rates was 0.68 within the tightening year and only slightly higher after a further year (0.70). Pass-through of the easing ranges from 0.63 to 0.95 depending on the window.
- *Outstanding loans lag.* Pass-through was 0.39 in T1, rising to 0.44 in T2. In the easing it was 0.69–0.78, higher than in the tightening. This reflects the larger share of loans linked to an external benchmark (mostly the repo), which reached 67.6% of floating-rate loans by March 2026 and reprice almost immediately.
- *Deposit rates lag most and depend most on the window.* Pass-through to outstanding deposits was 0.48 in T1 but 0.77 in T2. Most deposit repricing happened in the year *after* the last hike, as fixed-rate term deposits rolled over. In the easing it was 0.31–0.49.
- *Lags matter.* A contemporaneous window understates deposit pass-through by about 30 pp. Windows that start after the first cut (E1 versus E2) produce very different ratios, because rates had already begun adjusting in anticipation.

(b) Distributed-lag estimates

We estimate $\Delta R_t = \alpha + \beta_0 \Delta i_t + \beta_1 \Delta i_{t-1} + u_t$ on annual end-March changes for FY2015–16 to FY2025–26 ($T = 11$; $T = 9$ for MCLR). Choosing $p = 1$, one year, is justified by the results above, where deposit repricing extends about 12 months beyond the policy change. With 11 observations a longer lag cannot be estimated. Standard errors are OLS; the Durbin–Watson statistics of 1.7–2.8 show no strong autocorrelation.

Table 8: Distributed-lag pass-through (annual end-March changes, pp on pp). Asymmetry: tightening and easing terms entered separately; the test compares the two cumulative sums.

	WALR fresh	WALR outst.	1-yr MCLR	WADTDR outst.
Constant	−0.082	−0.146*	0.042	−0.021
	(0.072)	(0.078)	(0.091)	(0.071)
Δi_t (β_0)	0.670***	0.364***	0.475***	0.454***
	(0.069)	(0.074)	(0.082)	(0.068)
Δi_{t-1} (β_1)	0.145*	0.196**	0.202*	0.399***
	(0.071)	(0.076)	(0.085)	(0.070)
Observations	11	11	9	11
R^2 (adj.)	0.910	0.756	0.829	0.892
AIC	1.5	3.2	4.4	1.3
Cumulative $\beta_0 + \beta_1$	0.82	0.56	0.68	0.85
95% CI	[0.63, 1.00]	[0.36, 0.76]	[0.45, 0.90]	[0.67, 1.04]
Cum. tightening	0.71	0.60	0.44	0.70
Cum. easing	0.96	0.51	1.02	1.06
Asymmetry test p	0.461	0.783	0.077	0.271
Durbin–Watson	2.31	2.75	2.74	1.69

Results.

- *Fresh loans.* The impact effect dominates: $\hat{\beta}_0 = 0.67$ (0.07) against $\hat{\beta}_1 = 0.15$ (0.07). Cumulative pass-through is 0.82, with a 95% CI of $[0.63, 1.00]$. The strongest response comes within the same year.
- *Outstanding loans.* Pass-through is slower and smaller: 0.36 on impact, 0.20 after a year, 0.56 cumulatively (CI $[0.36, 0.76]$). This reflects the stock of MCLR-linked and fixed-rate loans.
- *MCLR.* 0.48 on impact plus 0.20 after a year, 0.68 cumulatively.
- *Deposits.* The lagged response is as large as the impact response: $\hat{\beta}_0 = 0.45$, $\hat{\beta}_1 = 0.40$. Cumulative pass-through is 0.85 (CI $[0.67, 1.04]$), statistically indistinguishable from full pass-through within two years.

Deposit rates thus catch up with lending rates over time, which compresses banks' margins during tightening and then rebuilds them.

Asymmetry. Point estimates suggest easing passes through *more* than tightening: 0.96 vs 0.71 for fresh loans, 1.06 vs 0.70 for deposits, and 1.02 vs 0.44 for MCLR. This partly reflects the deeper external-benchmark share in 2025. However, equality is not rejected at 5% for any rate; the smallest p-value, 0.08, is for MCLR. With 11 annual observations these tests have little power, so the data neither confirm nor rule out an asymmetry.

(c) Lending rates and bank credit

Data. Growth of non-food bank credit (end-March, %) is taken from *Handbook* Table 41, excluding the July 2023 merger of a non-bank with a bank. Credit growth was 15.4% in FY23, 16.5% in FY24, 12.1% in FY25

and 16.9% in FY26.

Lead–lag analysis. Credit growth is *positively* correlated with the change in the fresh-loan rate in the same year (0.51) and the previous year (0.51). It is uncorrelated with next year's change (-0.06). A regression of credit growth on ΔWALR_t, ΔWALR_{t-1} and real GDP growth gives positive coefficients of 1.9 and 2.1 ($T = 10$; Table 9). The residuals are autocorrelated (Breusch–Godfrey $p = 0.04$), so the table follows our convention and reports HAC(1) standard errors (1.18 and 0.70). On these the lagged coefficient is significant. With only 10 observations, however, HAC standard errors are themselves biased downward. With plain OLS standard errors (about 1.4) neither coefficient is significant. The robust finding is the *sign*: higher lending rates go with faster, not slower, credit growth.

Table 9: Credit growth and lending-rate changes (annual, FY2016–17 to FY2025–26). HAC(1) s.e.

	WALR fresh	WALR outst.
Constant	10.192***	9.708***
	(0.748)	(1.140)
ΔWALR_t	1.880	
	(1.181)	
ΔWALR_{t-1}	2.115***	
	(0.695)	
ΔWALR_t^O		1.683
		(2.074)
$\Delta\text{WALR}_{t-1}^O$		3.433***
		(1.072)
Real GDP growth$_t$	0.324**	0.457***
	(0.161)	(0.173)
Observations	10	10
R^2 (adj.)	0.370	0.321
AIC	55.0	55.8
Breusch–Godfrey(1) p	0.040	0.020
Durbin–Watson	1.84	2.39

Lending rates do not *lead* credit in the direction a transmission story would predict, where higher rates are followed by slower credit. Instead, rates and credit move together. That pattern is the signature of credit *demand* shocks. Credit growth itself does not reject a unit root (ADF $p = 0.60$, Table 15), though with 10 observations the test has almost no power. In FY23–24 strong demand for loans pushed up both quantities and prices, and banks financed it by raising deposit rates. Several mechanisms explain why the correlation need not reveal the causal effect of policy:

1. *Deposit repricing.* Banks' cost of funds adjusts with a lag (part (b)), so during tightening banks' margins on new loans are initially squeezed, delaying the rise in lending rates.

2. *Bank liquidity.* The banking system ran a liquidity surplus in 2021–22 and a deficit in 2023–24. The RBI's 2025 measures, including cash reserve ratio (CRR) cuts from 4.0% to 3.0% (Table 40), shifted credit supply independently of the repo rate.

3. *Loan contracts.* Repo-linked loans reprice immediately, MCLR-linked loans at reset dates, and fixed-rate loans not at all. The average outstanding rate is therefore a slow-moving mixture.

4. *Credit demand.* Strong growth, the post-pandemic investment cycle and retail credit expansion raise both credit and rates at the same time. The macroprudential tightening on unsecured consumer credit in November 2023 cut credit growth in FY25 independently of rates.

Identifying causal transmission would require bank- or loan-level variation in exposure to policy, such as differences in external-benchmark shares or deposit franchises across banks, or exogenous policy shocks. Annual aggregate correlations cannot provide it.

4 Q4. Fiscal policy: cycle versus choice

(a) Deficits, composition and the fiscal stance

Data. Central government accounts come from RBI *Handbook* Tables 86, 87 and 89 (sourced from Budget documents): actuals to 2024–25, RE for 2025–26 and BE for 2026–27. The ratios use nominal GDP on the old base up to 2021–22 and the new base from 2022–23. For 2026–27 we apply the Budget's 10% nominal growth assumption to the 2025–26 provisional estimate.

The new base lowers 2025–26 GDP by about 3%. Our ratios for 2025–26 and 2026–27 (4.50% and 4.45%) are therefore slightly higher than the official 4.4% and 4.3%, which used the GDP assumed at Budget time.

Sign convention.

- The primary balance is $PB = -$(gross primary deficit)/GDP, so a surplus is positive.
- The cyclically adjusted primary balance is $CAPB_t = PB_t - \varepsilon\,\tilde{y}_t$, where $\tilde{y}_t$ is the fiscal-year average of the quarterly HP gap.
- We set $\varepsilon = 0.10$. The Centre's net revenue receipts are about 9% of GDP with a buoyancy near 1.1, and expenditure is assumed not to respond to the cycle. Since states receive a large share of gross taxes, the Centre's semi-elasticity is lower than for general government. We report $\varepsilon = 0.20$ as a sensitivity.
- The fiscal impulse is $FI_t = -\Delta CAPB_t$, so a positive value means the stance is expansionary.

Table 10: Central government: deficits and spending (% of GDP), output gap, CAPB and fiscal impulse (FI, pp of GDP; + = expansionary).

FY	GFD	GPD	RD	CapEx	RevEx	Gap	CAPB	FI(ε=.1)	FI(ε=.2)
2014-15	4.10	0.87	2.93	1.58	11.77	-1.08	-0.76	-0.25	-0.23
2015-16	3.87	0.66	2.49	1.84	11.17	-0.37	-0.62	-0.14	-0.07
2016-17	3.48	0.36	2.06	1.85	10.98	0.88	-0.45	-0.18	-0.05
2017-18	3.46	0.36	2.60	1.54	10.99	1.27	-0.49	0.05	0.08
2018-19	3.44	0.35	2.40	1.63	10.62	2.09	-0.56	0.07	0.15
2019-20	4.64	1.60	3.32	1.67	11.69	1.20	-1.72	1.16	1.07
2020-21	9.16	5.73	7.30	2.15	15.53	-10.01	-4.73	3.01	1.89
2021-22	6.72	3.30	4.37	2.51	13.57	-4.62	-2.84	-1.89	-1.35
2022-23	6.65	3.10	4.10	2.83	13.22	-1.99	-2.90	0.06	0.32
2023-24	5.71	2.04	2.64	3.27	12.06	-0.61	-1.98	-0.92	-0.78
2024-25	4.95	1.44	1.77	3.31	11.32	0.15	-1.46	-0.52	-0.44
2025-26RE	4.50	0.82	1.52	3.16	11.17	1.16	-0.94	-0.52	-0.42
2026-27^{BE}	4.45	0.77	1.55	3.21	10.83	1.23	-0.89	-0.05	-0.04

GFD/GPD/RD: gross fiscal / gross primary / revenue deficit. RE: revised estimates; BE: budget estimates. 2022–23 onwards uses new-base GDP (the official 2022–23 GFD on the old base is 6.4%). The 2026–27 output gap is the 2026Q2 value only (one quarter of the fiscal year), so the 2026–27 CAPB and impulse are provisional as well as budget-based.

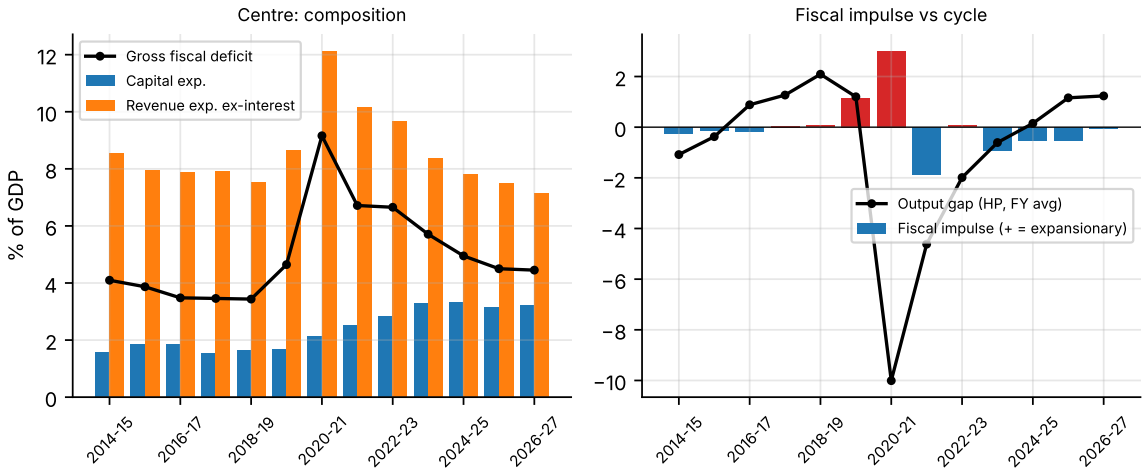


Figure 5: Left: Centre's capital expenditure, revenue expenditure net of interest, and gross fiscal deficit (% of GDP). Right: fiscal impulse against the output gap.

Deficits.

- The fiscal deficit rose from 3.4% of GDP in 2018–19 to 4.6% in 2019–20, already above target before the pandemic, then to 9.2% in 2020–21.
- It has since fallen in every year: 6.7% (2021–22), 6.65% (2022–23), 5.7%, 4.95%, 4.5% (2025–26 RE) and 4.45% (2026–27 BE).
- The primary deficit fell from 5.7% to 0.8% of GDP.

Composition. Capital expenditure doubled as a share of GDP, from 1.6% (2014–15) to a plateau of 3.2–3.3% since 2023–24. Revenue expenditure fell from a pandemic peak of 15.5% to 10.8%, below its pre-pandemic level of about 11%. Consolidation came entirely from revenue spending while capital spending was protected, which matches the policy emphasis in the background note.

Stance.

- *Pandemic (2019–20 to 2020–21).* The impulse was $+1.2$ and $+3.0$ pp of GDP while the gap fell to -10%. This was strongly counter-cyclical, although the 2019–20 impulse was largely a revenue shortfall and corporate tax cut rather than planned stimulus.
- *Early consolidation (2021–22).* The impulse was -1.9 pp with the gap still -4.6%. Withdrawing support before the gap had closed was *pro-cyclical*, although part of it was the reversal of one-off pandemic spending.
- *2022–23 to 2024–25.* Neutral (0.1), then -0.9 and -0.5 pp as the gap closed. Consolidation was gradual and timed with the recovery.
- *2025–26.* -0.5 pp with a gap of $+1.2\%$, which is counter-cyclical consolidation. In 2026–27 (BE) the impulse is about zero.

Over 2015–16 to 2026–27 the impulse is negatively correlated with the level of the gap (-0.45) and very strongly with its change (-0.93). The fiscal stance has been broadly counter-cyclical, with the 2021–22 withdrawal the main exception. The conclusions are unchanged with $\varepsilon = 0.20$ or the Hamilton gap. The 2025–26 impulse is -0.42 with $\varepsilon = 0.20$ and -0.63 with the Hamilton gap. The pandemic impulses are $+1.0$ and $+2.8$ pp with the Hamilton gap, against $+1.2$ and $+3.0$ in the baseline, and the 2021–22 withdrawal is -1.9 pp under both.

(b) Expenditure and growth

We estimate $g_t = \alpha + \rho g_{t-1} + \beta_K \Delta k_t + \beta_R \Delta r_t + \gamma' X_t + u_t$ on fiscal years 2006–07 to 2025–26 ($T = 20$). The variables are:

- g_t: real GDP growth (spliced series, %);
- $\Delta k_t = (K_t - K_{t-1})/Y_{t-1} \times 100$: the change in Centre capital expenditure as a share of *last year's* nominal GDP;
- Δr_t: the same for revenue expenditure net of interest payments.

Both spending terms are measured in pp of GDP, so they are comparably scaled, and they avoid the mechanical denominator link in $\Delta(K/Y)$. Interest is excluded because it is a transfer to bondholders rather than demand for goods. X_t contains dummies for 2020–21 and 2021–22.

Table 11: Expenditure–growth regressions. Dependent variable: real GDP growth (%). HAC(1) s.e.

	(1)	(2)	(3) + lags	(4) lags only
Constant	6.896***	5.555***	4.005***	4.418***
	(1.603)	(1.248)	(1.283)	(1.449)
g_{t-1}	0.193	0.093	0.389**	0.463***
	(0.169)	(0.183)	(0.161)	(0.154)
Δk_t	1.347	3.075***	2.492***	
	(1.590)	(0.771)	(0.912)	
Δr_t	−1.795	0.088	−0.173	
	(1.272)	(0.381)	(0.606)	
Δk_{t-1}			−1.635***	−2.415**
			(0.590)	(1.064)
Δr_{t-1}			0.199	−0.020
			(0.565)	(0.432)
FY2020-21 dummy		−13.421***	−11.947***	−11.587***
		(1.573)	(1.912)	(0.812)
FY2021-22 dummy		2.020	5.852***	9.136***
		(2.604)	(1.832)	(2.587)
Observations	20	20	20	20
R^2 (adj.)	0.251	0.825	0.851	0.763
AIC	103.8	76.1	73.7	82.1
Wald $\beta_K = \beta_R$ (p)	0.017	0.001	0.333	–
Cum. capital ($\Sigma\beta_K$)	–	–	0.86 (1.03)	–
Cum. revenue ($\Sigma\beta_R$)	–	–	0.03 (0.31)	–
Breusch–Godfrey(1) p	0.103	0.425	0.241	0.494

Results (column 2). An increase in capital spending equal to 1% of GDP is associated with 3.1 pp higher

growth in the same year ($\hat{\beta}_K = 3.08$, s.e. 0.77). Revenue spending has no association ($\hat{\beta}_R = 0.09$, s.e. 0.38). Equality is rejected (Wald $p = 0.001$).

Scaled by typical variation, a one standard deviation change in Δk (0.40 pp) is associated with 1.2 pp of growth, while one in Δr (1.05 pp) is associated with about 0.1 pp. Without the pandemic dummies (column 1) both coefficients are imprecise and revenue spending turns negative. This is the counter-cyclical pandemic transfer surge: revenue spending rose exactly when growth collapsed.

Diagnostics. No column shows residual autocorrelation (Breusch–Godfrey $p \geq 0.10$, Table 11). Annual growth and Δk_t reject a unit root at 5% ($p = 0.011$ and 0.015), whereas Δr_t does not ($p = 0.30$; Table 15). The failure for Δr_t comes from the single 2020–21 spike and its reversal, which the dummies absorb in the regression. With 20 observations these tests should be read as indicative only.

(c) Lags and cumulative associations

Results (column 3). Adding Δk_{t-1} and Δr_{t-1} reveals a reversal for capital spending:

- the impact coefficient is 2.49 (0.91), but the lag is -1.64 (0.59);
- the cumulative association is 0.86 (s.e. 1.03, 95% CI $[-1.17, 2.88]$);
- for revenue spending the cumulative association is 0.03 (0.31);
- the cumulative difference is not significant ($p = 0.33$).

Capital spending is therefore associated with growth *contemporaneously*, not with a longer delay. Revenue spending shows no association at any horizon. The lagged-dependent-variable model passes a Breusch–Godfrey test ($p = 0.24$).

Why these are not fiscal multipliers.

1. *Reverse causality.* Capital expenditure is the residual item that adjusts to revenue during the year. In fast-growth years tax buoyancy lets the Centre spend its capital budget, and in shortfall years capital spending is cut first. Growth therefore causes Δk, which inflates $\hat{\beta}_K$ and is consistent with the sign reversal at the lag.
2. *Common shocks.* The pandemic, commodity cycles and the global cycle move growth and the budget together. Two dummies cannot absorb all of this.
3. *Scope.* The Centre's spending is about half of general government spending. State capital expenditure and public enterprise investment are omitted.
4. *Missing policy interaction.* A multiplier depends on the monetary response and the state of the cycle, which a single-equation regression does not condition on.
5. *Small sample.* With $T = 20$ the confidence intervals are wide.

Credible multipliers need exogenous variation, for example from narrative identification of spending decisions, SVARs with institutional timing restrictions (Blanchard–Perotti), or cross-state variation in central transfers.

5 Q5. Households and the 2025 policy mix

(a) Consumption and income

Measurement. India does not publish quarterly household disposable income; the household sector is a residual in the national accounts. We therefore use real GDP as the income proxy in quarterly regressions (2005Q3–2026Q2). For annual levels we use the change in real PFCE (Rs lakh crore) on the change in real GDP, which gives a marginal propensity to consume out of output. Gross national disposable income (GNDI) is available on a consistent basis only from 2020–21 in the current *Handbook* table (six annual observations), which is too short to estimate a two-lag model. The problem statement also allows an alternative based on saving, but household saving is published only annually and with a long lag. GDP is therefore the only income measure that covers the 2025 episode at a usable frequency. We correct for this in the interpretation below.

- ΔC_t, ΔY_t: quarter-on-quarter log changes of SA real PFCE and real GDP (%).
- Z_t: COVID dummies for 2020Q2–Q3 and 2021Q2–Q3, plus the lagged change in the repo rate.

Table 12: Consumption–income regressions (HAC(4) s.e.).

	(1) q/q	(2) q/q + rate	(3) q/q ex-COVID	(4) y/y
Constant	0.486	0.444	0.634	1.520
	(0.319)	(0.427)	(0.582)	(0.978)
ΔY_t	0.634***	0.596**	0.622***	
	(0.154)	(0.281)	(0.189)	
ΔY_{t-1}	0.092***	0.116*	0.014	
	(0.027)	(0.060)	(0.163)	
Δi_{t-1}		0.476		
		(0.915)		
$\Delta_4 Y_t$				0.830***
				(0.092)
$\Delta_4 Y_{t-4}$				−0.043
				(0.063)
Observations	84	50	76	81
R^2 (adj.)	0.525	0.688	0.080	0.759
AIC	375.9	215.2	338.2	366.9
$\beta_0 + \beta_1$	0.73	0.71	0.64	0.79
95% CI	[0.44, 1.01]	[0.26, 1.16]	[0.00, 1.27]	[0.53, 1.04]
COVID dummies	Yes	Yes	Sample excl.	No
Breusch–Godfrey(4) p	0.000	0.004	0.000	0.006

Table 13: Annual levels: marginal propensity to consume out of real GDP, FY2006–07 to FY2025–26 (HAC(1) s.e.).

	ΔC_t (Rs lakh crore)
Constant	0.947^{***}
	(0.291)
ΔY_t	0.260^{***}
	(0.039)
ΔY_{t-1}	0.219^{***}
	(0.039)
FY2020-21	-6.288^{***}
	(0.908)
FY2021-22	10.982^{***}
	(1.318)
Observations	20
R^2 (adj.)	0.980
AIC	41.0
$\beta_0 + \beta_1$	0.48 (0.02)
Average C/Y, FY16–26	0.56

Quarterly results. The contemporaneous elasticity is 0.63 (0.15) and the lagged one 0.09 (0.03), a combined 0.73 (95% CI $[0.44, 1.01]$). Excluding the pandemic gives 0.64. The year-on-year version gives 0.79. Consumption responds less than one-for-one to income within a year, and almost the whole response occurs in the same quarter. The lagged change in the repo rate has no detectable effect (0.48, s.e. 0.92).

Annual results. The marginal propensity to consume out of GDP is 0.26 in the current year plus 0.22 in the next, 0.48 in total (s.e. 0.02). The two frequencies agree. Converting the quarterly elasticity to a marginal propensity (elasticity $\times$ average C/Y, 0.73×0.56) gives about 0.41, close to the annual 0.48.

Interpretation. Two features of the estimates are informative. The fact that the MPC (0.48) is below the APC (0.56) is *not* one of them: a Keynesian consumption function with an intercept implies the same thing. What matters is the following:

- *Size of the response.* The elasticity is below one (0.73, upper CI bound 1.01), so the consumption share falls when output rises. Households save part of each income gain, as they would if they treat part of it as transitory.
- *Timing.* Only about 40% of the annual response (0.22 of 0.48) comes with a one-year lag. Under strong smoothing the lagged response would dominate; under hand-to-mouth behaviour the response would be immediate and larger. The estimates sit between the two.

Caveats.

- GDP includes corporate and government income, so the true propensity out of household disposable income is higher.
- Quarterly PFCE is noisy and partly estimated from commodity flows, its quarter-on-quarter volatility is about 2.9 times that of GDP (excluding the pandemic). This biases the quarterly elasticities toward zero.
- The 2022 base revision is handled by the splice but not removed.
- The quarterly residuals are autocorrelated (Breusch–Godfrey $p < 0.01$ in every column of Table 12), as expected with noisy PFCE. HAC standard errors account for this. Both growth series are stationary (ADF $p < 0.001$).

The evidence is consistent with *moderate* smoothing, not with a strong response to current income. It cannot distinguish credit-constrained households from forward-looking ones.

(b) Taxes and consumption

We regress annual real PFCE growth on changes in the Centre's personal income tax (PIT) take as a share of GDP (net of the states' share, Table 87), its lag, real GDP growth and pandemic dummies (FY2006–07 to FY2025–26, $T = 20$). Columns 3–4 add indirect taxes or replace PIT with total direct taxes. Column 5 adds a narrative dummy for the three fiscal years with a major household-facing tax cut: 2019–20 (the Rs 5 lakh rebate), 2023–24 (the new regime as default, with a Rs 7 lakh rebate) and 2025–26 (the Rs 12 lakh threshold plus GST 2.0).

A caveat applies to all columns. A change in a tax-to-GDP ratio is not the same as a change in tax *policy*. The ratio also moves with tax buoyancy, bracket creep, compliance and the composition of growth, all of which are correlated with consumption. The ratio regressions therefore mix policy and cyclical effects. The narrative dummy isolates policy decisions, but only three of them.

Table 14: Tax changes and consumption growth (HAC(1) s.e.).

	(1)	(2)	(3)	(4)	(5)
Constant	7.384***	4.432***	3.998***	4.067***	4.802***
	(0.261)	(0.946)	(0.807)	(0.843)	(0.878)
$\Delta(T^{PIT}/Y)_t$	1.339	−0.177	−0.602		−0.320
	(1.630)	(0.863)	(0.837)		(0.704)
$\Delta(T^{PIT}/Y)_{t-1}$	−1.930	−0.946			−0.608
	(1.212)	(1.605)			(1.491)
$\Delta(T^{DIR}/Y)_t$				−0.065	
				(0.442)	
$\Delta(T^{IND}/Y)_t$			−0.425	−0.397	
			(0.422)	(0.455)	
Real GDP growth$_t$		0.406***	0.454***	0.441***	0.372***
		(0.116)	(0.101)	(0.105)	(0.105)
Tax-relief year dummy					−0.862**
					(0.349)
FY2020-21 dummy	−12.953***	−7.490***	−6.354***	−6.548***	−8.011***
	(0.311)	(1.678)	(1.504)	(1.523)	(1.542)
FY2021-22 dummy	3.930***	3.392***	3.351***	3.273***	3.388***
	(0.536)	(0.421)	(0.441)	(0.417)	(0.344)
Observations	20	20	20	20	20
R^2 (adj.)	0.866	0.908	0.908	0.907	0.914
AIC	67.6	60.6	60.7	60.9	59.9
Breusch–Godfrey(1) p	0.545	0.556	0.719	0.809	0.309

Results. Without controlling for output (column 1) the PIT coefficients have opposite signs and are insignificant. Controlling for growth (column 2), a 1 pp of GDP rise in PIT is associated with 0.18 pp lower consumption growth on impact and 0.95 pp lower a year later, a cumulative -1.12 (s.e. 1.53, 95% CI $[-4.1, 1.9]$). Indirect taxes have the expected negative sign (-0.42, s.e. 0.42) but are also insignificant.

The signs match Keynesian behaviour, but we cannot reject zero, the Ricardian benchmark, at conventional levels. Equally, the confidence intervals include large Keynesian effects. The data cannot discriminate.

Narrative tax cuts (column 5). The tax-relief dummy is *negative* and significant: consumption growth was 0.86 pp lower (s.e. 0.35) in tax-relief years, after controlling for GDP growth. Taken literally this contradicts both a Keynesian response (which predicts higher consumption) and Ricardian equivalence (which predicts no change). The likelier explanation is that the timing of relief is endogenous. Governments cut taxes when household demand is weak: the 2019 rebate came during the 2019 consumption slowdown, and the 2025 relief followed weak urban consumption in 2024–25. That reverse causality biases the coefficient downward. With three events and no exogenous timing, column 5 is evidence of endogenous policy rather than of the household response. No column shows residual autocorrelation (Breusch–Godfrey $p \geq 0.31$).

Reasons for weak identification.

- *Narrow base.* The income-tax base is narrow (PIT is only about 2.4% of GDP), so changes in the PIT ratio are small.
- *2025 tax cuts.* The 2025–26 Budget raised the zero-tax threshold under the new regime to Rs 12 lakh, yet PIT/GDP fell only from 2.45% to 2.35% (Table 87, RE). Applying the column-2 cumulative coefficient implies at most about 0.1 pp extra consumption growth. The larger household-facing change came on the indirect side. GST 2.0 (effective 22 September 2025) cut rates on many everyday goods. The Centre's indirect taxes fall from 3.43% of GDP in 2024–25 to 3.36% in 2025–26 (RE, half a year of the new rates) and 3.05% in 2026–27 (BE, a full year), a cumulative fall of about 0.4 pp of GDP. That is four times the fall in the PIT ratio. Our annual data end in 2025–26, so they capture only the first half-year of this change.
- *Who pays.* The taxpayers who benefit are mostly in upper-income deciles with low marginal propensities to consume and good access to credit.

Interpretation.

- *Credit constraints and heterogeneity.* The majority of Indian households who are credit constrained or hand-to-mouth, such as informal workers and rural households, pay little income tax. For them the relevant fiscal instruments are indirect taxes and transfers. A Keynesian response to income-tax cuts is therefore unlikely to be large in aggregate even if constrained households spend most of any windfall. The 2025 GST cuts reach these households more directly. Their effect on consumption should appear in 2025–26 Q3 to 2026–27 data, which is the natural test of the Keynesian channel once a full year of national-accounts data is available.
- *Expectations of future taxation.* The government has explicitly anchored its debt path (debt-to-GDP of $50 \pm 1\%$ by 2030–31). Debt-financed tax cuts may therefore be read as temporary, which pushes behaviour toward the Ricardian end.
- *Aggregation.* Full Ricardian equivalence also requires lump-sum taxes and infinite horizons, neither of which holds. The weak aggregate response more likely reflects incidence on high-income, low-MPC households than forward-looking offsetting saving.

Overall, the evidence is "not inconsistent with" Ricardian behaviour but more plausibly reflects heterogeneity.

(c) Was the 2025 mix complementary or offsetting?

Definitions. The two terms can be judged on two levels, and we keep them separate.

- *Demand effects.* In 2025 monetary easing added to demand while fiscal consolidation subtracted from it. On this level the two instruments were *offsetting* by construction, and the question is only which was larger.
- *Objectives.* The policies are complementary if, together, they moved the economy towards both targets (inflation near 4% with output near potential, and a falling debt path) at lower cost than either could alone.

Our verdict uses the second level, and we report the first alongside it.

Evidence.

1. **Slack (Q1).** The 2025 gap was $+0.7\%$ (HP) and $+0.6\%$ (broken trend), essentially closed, while inflation averaged 2.2%. Demand was not weak. The disinflation came from food prices and, from September, from the GST cut.
2. **Rule (Q2).** The repo averaged 2.6 pp *above* the Rule-A benchmark in 2025 (5.8% vs 3.2%). Easing was justified by the benchmark but executed cautiously. Against the RBI's own estimated reaction function, the cuts were if anything slightly faster than usual: a dynamic simulation puts the repo at 5.62% in 2025Q4, against an actual 5.43%.
3. **Transmission (Q3).** By August 2026 the cuts had passed through 0.63 to fresh lending rates and 0.74 to outstanding loans, but only 0.37–0.47 to deposit rates. Deposit rates cumulate to near-full pass-through only after about two years. Easing therefore lowered borrowing costs faster than it eroded saving returns, which supports demand.
4. **Fiscal stance (Q4).** The impulse was -0.5 pp of GDP in 2025–26 (about 0 in 2026–27 BE), but with capital expenditure held at about 3.2% of GDP. Capital spending is the component most associated with growth on impact ($\hat{\beta}_K = 3.1$, s.e. 0.8). That association reverses a year later, however: the cumulative coefficient is 0.86 with a 95% CI of $[-1.2, 2.9]$, and it is probably inflated by reverse causality (Q4(c)). Protecting capital spending is therefore a reasonable reading of the evidence, not a measured multiplier.

Assessment. On demand the instruments offset each other: monetary easing roughly offset a fiscal drag of about 0.5% of GDP. On objectives the mix was *complementary*. Growth stayed above 7% and the deficit fell to 4.5% of GDP while inflation, pushed below target by supply factors and a tax cut, was not reinforced by a demand squeeze. Each instrument did the job it is better suited to: monetary policy handled the cyclical inflation undershoot, and fiscal policy rebuilt buffers while protecting capital spending.

A further channel is plausible but untested here. Lower central borrowing should reduce pressure on government bond yields and bank funding, and so help the repo cuts transmit. We do not have government bond yield data in our sample, so we present this as a hypothesis rather than a finding. The faster pass-through to outstanding loans in the easing (0.69–0.78, against 0.39–0.44 in the tightening) is consistent with it, but it is equally explained by the rise in external-benchmark lending.

Recommendation for 2026–27. Inflation has returned to 5.5% (September 2026) with a gap of about $+1.2\%$, and Rule A prescribes about 7.5% against a repo of 5.50%. We recommend the following:

- The RBI should continue *calibrated* tightening of a further 25–50 bp, conditional on core and food inflation. Given our transmission estimates, about 0.7 pass-through to new loans within a year, the effect on borrowing costs will be rapid.
- The Centre should hold the 2026–27 BE deficit of 4.3% (official basis) and avoid pro-cyclical slippage, while protecting the capital expenditure share.
- Further income-tax relief would do little for demand (Q5(b)) and would weaken the debt anchor.

Alternative interpretation. On the Hamilton filter the economy has been 4.5–5% *above* potential since 2023, and still about 3% above it when the pandemic quarters are excluded from the estimation. On that reading the 2025 easing was pro-cyclical stimulus to an overheated economy and helped cause the 2026 inflation rebound. A further possibility is that the 2026 rebound is itself a supply shock: food prices normalising, the GST cut dropping out of year-on-year inflation from September 2026, and the rupee near Rs 95/US$ in October 2026. In that case further hikes would cost output for little gain.

Main uncertainty. The *size of the output gap*, together with the neutral rate. Our gap estimates span about 5 pp for 2025 (-1.8% to $+4.6\%$, or -1.8% to $+2.8\%$ once the Hamilton regression excludes the pandemic), are revised by 0.5–4 pp at the end of the sample, and rest on a newly rebased GDP series. If true slack is near the Hamilton estimate, the conclusion changes from "complementary and appropriately cautious" to "too loose". Better real-time measures of slack, such as capacity utilisation and labour-market data, would matter more than any refinement of the rule.

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A Data log

Variable	Units / transformation	Source	Status / notes
Real GDP, nominal GDP, real PFCE (quarterly)	Rs million, NSA → ratio-spliced, SA (STL / quarter effects), 100 log	IMF IFS via FRED: NGDPRNSAXDCINQ, NGDPNSAXDCINQ, NCPRNSAXDCINQ (updated 7 Sep 2026)	2004Q2–2026Q2. New base from 2022Q2; 2025–26 provisional, 2026Q2 first estimate. Splice uses *Handbook* Tables 1–2 (2022–23 in both bases).
Annual GDP, GNDI, PFCE per capita	Rs crore	RBI *Handbook* 2025–26, Tables 1–2 (NSO)	2024–25 first revised; 2025–26 provisional.
CPI-Combined inflation	% y/y, monthly; quarterly mean	MoSPI/PIB release 12 Feb 2025 (Annex V–VI); *Handbook* Table 243; MPC statement Oct 2026; MoSPI Sep 2026 release	Apr–May 2020 computed from imputed indices. Base 2024=100 from Jan 2026. Sep 2026 provisional.
Policy repo rate	%, daily step → quarter mean / end	*Handbook* Table 40; RBI MPC resolution 7 Oct 2026	Actual.
WALR (fresh, outstanding), 1-yr MCLR, WADTDR	%, end-March	*Handbook* Table 59	2026–27 column = May 2026.
WALR, WADTDR monthly	%	RBI press releases "Lending and Deposit Rates of SCBs" (Apr 2025–Sep 2026)	Feb 2025–Aug 2026; months not retrieved left blank.
Non-food bank credit	Rs crore, end-March; % growth	*Handbook* Table 41	FY24–25 growth excludes the July 2023 merger.
GFD, GPD, RD; revenue & capital expenditure; interest; PIT, direct, indirect tax	Rs crore; % of GDP	*Handbook* Tables 86, 87, 89 (Budget documents)	Actuals to 2024–25; 2025–26 RE; 2026–27 BE (nominal GDP = 1.10 × 2025–26).

Table 15: Augmented Dickey–Fuller tests (constant, no trend). Quarterly series: lag length by AIC. Annual series: one lag, since AIC selection is unreliable with 20 or fewer observations. H_0: unit root.

Series	Freq.	Sample	ADF stat.	p-value	Lags	Obs.
Inflation π_t	Q	2014Q1–2026Q3	-1.93	0.317	4	46
Output gap (HP)	Q	2011Q1–2026Q2	-5.81	0.000	0	61
Output gap (Hamilton)	Q	2011Q1–2026Q2	-3.88	0.002	0	61
Output gap (Hamilton, ex-COVID est.)	Q	2011Q1–2026Q2	-2.13	0.232	8	53
Repo rate i_t	Q	2014Q1–2026Q2	-2.40	0.141	1	48
Δi_t	Q	2014Q2–2026Q2	-3.49	0.008	0	49
ΔWALR fresh	A	FY16–FY27	-2.43	0.135	1	10
ΔWALR outstanding	A	FY15–FY27	-2.67	0.079	1	11
ΔWADTDR outstanding	A	FY15–FY27	-3.45	0.009	1	11
Credit growth	A	FY16–FY26	-1.37	0.596	1	9
Real GDP growth g_t (annual)	A	FY07–FY26	-3.39	0.011	1	18
Δk_t	A	FY07–FY26	-3.29	0.015	1	18
Δr_t	A	FY07–FY26	-1.97	0.299	1	18
ΔC_t (q/q)	Q	2005Q3–2026Q2	-8.77	0.000	1	83
ΔY_t (q/q)	Q	2005Q3–2026Q2	-9.63	0.000	2	82
PFCE growth (annual)	A	FY07–FY26	-3.50	0.008	1	18
$\Delta(T^{PIT}/Y)$	A	FY07–FY26	-3.39	0.011	1	18

Annual tests have very low power; a non-rejection with 10–20 observations is weak evidence of a unit root.

Reproducibility. The three files in Appendix B reproduce every table and figure (`python analysis.py`; requires pandas, numpy, statsmodels, matplotlib). All raw numbers are typed into `data_inputs.py` with their sources.

Extending Q3(b) to monthly data. Download the WALR and WADTDR monthly series (2019 onwards) from RBI's Database on Indian Economy (https://data.rbi.org.in), replace `rates_A` with the monthly series, and set $p = 12$. The function used for Table 8 applies unchanged.

B Code

data_inputs.py

```
"""
ECO612 group assignment -- data inputs.
Every number below is transcribed from the source listed next to it (accessed 30 Oct 2026).
Units: crore = 10 million rupees; FRED/IMF series are in millions of rupees.
"""
import pandas as pd, numpy as np

# ---------------------------------------------------------------------------
# 1. Quarterly national accounts, IMF IFS via FRED (millions of INR, NSA).
# NGDPRNSAXDCINQ real GDP, NGDPNSAXDCINQ nominal GDP, NCPRNSAXDCINQ real PFCE.
# Dates are calendar-quarter starts; 2004-04-01 = Q1 of FY2004-05.
# From 2022-04-01 the IMF series is the new MoSPI base (2022-23); earlier
# values are an IMF-linked back series of the 2011-12 base (see splice below).
# ---------------------------------------------------------------------------
q_dates = pd.date_range("2004-04-01", "2026-04-01", freq="QS-JAN")
real_gdp =
    [18650362.7,18850258.9,21145497.6,21973763.5,20370646.8,20441921.1,23042974.4,24249816.4,
21879320.0,22534949.7,25226863.2,26626257.5,24244392.2,24594738.9,27887618.8,28976145.0,
26204314.2,26251284.5,28315604.9,29044540.3,27512912.4,28078026.8,30640610.6,32896403.1,
30344950.3,30795809.8,33916232.4,36293323.9,33424678.9,32471105.0,35368180.0,37598642.7,
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40305363.0,40726433.3,42056920.1,44247008.8,43365565.7,43998315.9,45083938.5,48268479.0,
47129611.2,48252868.9,48950108.8,51304241.8,50010528.1,50818927.3,52214433.8,55887203.2,
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```

```
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nom_gdp = [6953205.1,7185541.0,8108597.1,8409260.4,7914131.2,8087610.0,9230673.8,9690115.7,
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74031370.0,73677500.0,83548220.0,86728150.0,80001920.0,79861630.0,90833930.0,94670470.0,88268710.0]

real_pfce =
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19509008.1,19505855.5,21932872.7,21522476.2,21041213.8,20688095.1,23458901.9,23304937.0,
22566487.8,22497042.6,23958384.2,25129353.6,24171077.4,24094606.5,26010513.1,27339488.3,
25854213.3,26408748.0,28926344.8,28687861.0,28077034.4,27690540.5,30128277.4,30836340.3,
29847313.7,30068028.6,32379259.2,32713374.4,31989288.3,31916763.2,34316061.4,33247276.3,
24661785.0,29552710.0,34833611.5,35467264.2,29113201.8,33586163.6,38680192.1,37675168.5,
34478690.0,34254290.0,40372180.0,40283360.0,36892350.0,36242810.0,42597080.0,42980160.0,
39115680.0,38555690.0,45523270.0,45721180.0,41761840.0,41015610.0,49174270.0,49137350.0,44742780.0]

Q = pd.DataFrame({"rgdp":real_gdp,"ngdp":nom_gdp,"rpfce":real_pfce}, index=q_dates)

# ------------------------------------------------------------------------
# 2. Annual NAS benchmarks for the base-year splice (RBI HBS 2025-26 Tables 1-2, Rs crore).
# 2022-23 is published under BOTH bases, which identifies the splice ratio.
# ------------------------------------------------------------------------
NAS = dict(
   rgdp_old_FY22=15021846, rgdp_old_FY23=16164913, rgdp_new_FY23=26117627,
   rgdp_new={2023:26117627, 2024:28000767, 2025:29988619, 2026:32312034}, # FY ending March of
       year
   ngdp_old={2021:19854096, 2022:23597399, 2023:26890473},
   ngdp_new={2023:26117627, 2024:28983909, 2025:31807309, 2026:34635638},
   pfce_pc_old_FY22=63807, pfce_pc_old_FY23=67865, pfce_pc_new_FY23=107910, pop_FY22_lakh=13686,
   gndi_cur={2021:20139597, 2022:23925034, 2023:26566532, 2024:29455128, 2025:32451963,
       2026:35401638},
)

# ------------------------------------------------------------------------
# 3. CPI-Combined y/y inflation (%), monthly.
# 2014-01..2025-01: MoSPI/PIB release of 12 Feb 2025, Annexure VI (base 2012=100).
# Apr/May 2020 not released; computed from Annexure V imputed indices (151.4/141.2, 150.9/142.0).
# 2025-02..2025-12: RBI HBS 2025-26 Table 243 (base 2012). 2026-01..06: Table 243 (new base
    2024=100).
# 2026-07, 08: MPC resolution Oct 2026 (4.5, 4.8). 2026-09: MoSPI release (5.54, provisional).
# ------------------------------------------------------------------------
cpi_yoy = {
2014:[8.60,7.88,8.25,8.48,8.33,6.77,7.39,7.03,5.63,4.62,3.27,4.28],
2015:[5.19,5.37,5.25,4.87,5.01,5.40,3.69,3.74,4.41,5.00,5.41,5.61],
2016:[5.69,5.26,4.83,5.47,5.76,5.77,6.07,5.05,4.39,4.20,3.63,3.41],
2017:[3.17,3.65,3.89,2.99,2.18,1.46,2.36,3.28,3.28,3.58,4.88,5.21],
2018:[5.07,4.44,4.28,4.58,4.87,4.92,4.17,3.69,3.70,3.38,2.33,2.11],
2019:[1.97,2.57,2.86,2.99,3.05,3.18,3.15,3.28,3.99,4.62,5.54,7.35],
2020:[7.59,6.58,5.84,round((151.4/141.2-1)*100,2),round((150.9/142.0-1)*100,2)
    ,6.23,6.73,6.69,7.27,7.61,6.93,4.59],
2021:[4.06,5.03,5.52,4.23,6.30,6.26,5.59,5.30,4.35,4.48,4.91,5.66],
2022:[6.01,6.07,6.95,7.79,7.04,7.01,6.71,7.00,7.41,6.77,5.88,5.72],
2023:[6.52,6.44,5.66,4.70,4.31,4.87,7.44,6.83,5.02,4.87,5.55,5.69],
2024:[5.10,5.09,4.85,4.83,4.80,5.08,3.60,3.65,5.49,6.21,5.48,5.22],
```

```
2025:[4.31,3.6,3.3,3.2,2.8,2.1,1.6,2.1,1.4,0.3,0.7,1.3],
2026:[2.7,3.2,3.4,3.5,3.9,4.4,4.5,4.8,5.54],
}

# ------------------------------------------------------------------------
# 4. Policy repo rate (%), effective dates. RBI HBS 2025-26 Table 40; Oct-2026 hike from
# RBI MPC resolution of 7 Oct 2026 (5.25 -> 5.50).
# ------------------------------------------------------------------------
repo_changes = [("2013-01-29",7.75),("2013-03-19",7.50),("2013-05-03",7.25),("2013-09-20",7.50),
("2013-10-29",7.75),("2014-01-28",8.00),("2015-01-15",7.75),("2015-03-04",7.50),("2015-06-02"
    ,7.25),
("2015-09-29",6.75),("2016-04-05",6.50),("2016-10-04",6.25),("2017-08-02",6.00),("2018-06-06"
    ,6.25),
("2018-08-01",6.50),("2019-02-07",6.25),("2019-04-04",6.00),("2019-06-06",5.75),("2019-08-07"
    ,5.40),
("2019-10-04",5.15),("2020-03-27",4.40),("2020-05-22",4.00),("2022-05-04",4.40),("2022-06-08"
    ,4.90),
("2022-08-05",5.40),("2022-09-30",5.90),("2022-12-07",6.25),("2023-02-08",6.50),("2025-02-07"
    ,6.25),
("2025-04-09",6.00),("2025-06-06",5.50),("2025-12-05",5.25),("2026-10-07",5.50)]

# ------------------------------------------------------------------------
# 5. Bank lending/deposit rates, end-March (%). RBI HBS 2025-26 Table 59.
# FY label = year in which March falls; 2027 = "2026-27@" i.e. May-2026 values.
# Repo end-March from Table 40. Call rate end-March from Table 59 col.2.
# ------------------------------------------------------------------------
rates_A = pd.DataFrame({
 "walr_out": [12.11,11.76,11.20,10.80,10.25,10.23,9.92,9.10,8.74,9.72,9.83,9.77,8.99,8.97],
 "walr_fresh":[np.nan,11.07,10.47,9.74,9.34,9.69,8.71,7.92,7.63,9.32,9.37,9.35,8.40,8.51],
 "mclr_1y": [np.nan,np.nan,np.nan,8.60,8.40,8.75,8.20,7.30,7.25,8.55,8.80,9.00,8.40,8.65],
 "wadtdr_out":[8.79,8.57,7.73,6.97,6.67,6.89,6.38,5.28,5.03,6.22,6.96,7.11,6.62,6.57],
 "call": [8.28,7.97,6.98,6.25,5.94,6.27,5.43,3.44,3.26,5.42,6.63,6.51,5.45,5.24],
 "repo": [8.00,7.50,6.75,6.25,6.00,6.25,4.40,4.00,4.00,6.50,6.50,6.25,5.25,5.25],
}, index=range(2014,2028))

# Monthly WALR/WADTDR values published in RBI 'Lending and Deposit Rates of SCBs' press releases
# (rbi.org.in press releases 2025-26; latest revision used where revised). Gaps = not retrieved.
rates_M = pd.DataFrame({
 "walr_fresh":{"2025-02":9.40,"2025-03":9.35,"2025-04":9.26,"2025-05":9.20,"2025-06":8.62,"2025-12
     ":8.28,
            "2026-01":8.49,"2026-02":8.44,"2026-03":8.40,"2026-04":8.50,"2026-05":8.51,"2026-06"
                :8.53,
            "2026-07":8.52,"2026-08":8.61},
 "walr_out":{"2025-04":9.70,"2025-05":9.69,"2025-06":9.48,"2025-11":9.21,"2025-12":9.06,"2026-01"
     :9.04,
          "2026-02":9.00,"2026-03":8.99,"2026-04":8.98,"2026-05":8.97,"2026-06":8.96,"2026-07"
              :8.97,"2026-08":8.96},
 "wadtdr_fresh":{"2025-02":6.49,"2025-03":6.65,"2025-04":6.30,"2025-05":6.11,"2025-06":5.75,"
     2025-11":5.59,
             "2025-12":5.67,"2026-01":5.66,"2026-02":5.65,"2026-03":6.07,"2026-04":5.79,"2026-05"
                 :5.83,
             "2026-06":5.99,"2026-07":5.90},
 "wadtdr_out":{"2025-02":7.02,"2025-03":7.03,"2025-04":7.10,"2025-05":7.07,"2025-06":6.99,"2025-11
     ":6.73,
            "2025-12":6.68,"2026-01":6.64,"2026-02":6.62,"2026-03":6.62,"2026-04":6.59,"2026-05"
                :6.57,
            "2026-06":6.58,"2026-07":6.58,"2026-08":6.56},
})

# ------------------------------------------------------------------------
# 6. Scheduled commercial banks, end-March, Rs crore. RBI HBS 2025-26 Table 41.
# Non-food credit; 2023-24 and 2024-25 also published excluding the July-2023 HDFC merger.
# ------------------------------------------------------------------------
nonfood =
    {2013:5164037,2014:5895649,2015:6442002,2016:7144361,2017:7787539,2018:8583436,2019:9730112,
        2020:10319097,2021:10888255,2022:11836303,2023:13655330,2024:16409083,2025:18207441,2026:21289626}
```

```
nonfood_exmerger = {2024:15901477, 2025:17822605}

# ------------------------------------------------------------------------
# 7. Central government finances, Rs crore. RBI HBS 2025-26 Tables 86, 87, 89 (Budget documents).
# 2025-26 = Revised Estimates, 2026-27 = Budget Estimates; earlier years = Actuals.
# ------------------------------------------------------------------------
yrs = list(range(2005,2028)) # FY ending March
GFD = [125794,146435,142573,126912,336992,418482,373591,515990,490190,502858,510725,532791,535618,
     591062,649418,933651,1818291,1584521,1737755,1654643,1574431,1558492,1695768]
GPD =
    [-1140,13805,-7699,-44118,144788,205389,139569,242840,177020,128604,108281,91132,54904,62110,
     66770,321581,1138422,779022,809238,590771,458856,284154,291796]
RD =
    [78338,92300,80222,52569,253539,338998,252252,394348,364282,357048,365519,342736,316381,443600,

     454483,666545,1449599,1031021,1069926,765216,564296,526764,592344]
REVEX
    =[384329,439376,514609,594433,793798,911809,1040723,1145785,1243514,1371772,1466992,1537761,1690584,

     1878833,2007399,2350604,3083519,3200926,3453132,3494252,3600914,3869087,4125494]
INTEREST
    =[126934,132630,150272,171030,192204,213093,234022,273150,313170,374254,402444,441659,480714,
     528952,582648,612070,679869,805499,928517,1063872,1115575,1274338,1403972]
CAPEX
    =[81248,66362,68778,118238,90158,112678,156604,158580,166858,187675,196681,253022,284609,263140,

     307714,335726,426317,592875,740025,949195,1051953,1095755,1221821]
PIT =
    [35443,45238,62707,86563,86985,94532,102441,118224,140438,169408,188336,172748,225214,258461,
     303508,298204,299689,427228,529416,657366,778800,814513,922847]
DIRTAX
    =[95944,120692,169738,231574,248152,271623,313501,343310,396585,455829,500531,449296,521287,606216,

     723492,638365,583212,865385,1052856,1234173,1407508,1509493,1705812]
INDTAX
    =[128854,149572,181444,207972,195169,184913,256367,286454,345292,360025,403085,494469,580085,636272,

     593719,718537,843075,939408,1044930,1093077,1092530,1165168,1161110]
FISC = pd.DataFrame(dict(GFD=GFD,GPD=GPD,RD=RD,REVEX=REVEX,INT=INTEREST,CAPEX=CAPEX,PIT=PIT,DIR=
    DIRTAX,IND=INDTAX), index=yrs)
FISC_STATUS = {2026:"RE", 2027:"BE"}
NOMINAL_GROWTH_BE_2027 = 0.10 # Union Budget 2026-27 assumption (10% over FY26)
```

build.py

```
"""Construct analysis datasets (quarterly and annual) from data_inputs."""
import numpy as np, pandas as pd
import data_inputs as D

def fy_of(ts): # Indian fiscal year label = year in which March falls
   return ts.year + 1 if ts.month >= 4 else ts.year

def splice_quarterly():
   Q = D.Q.copy()
   Q["fy"] = [fy_of(t) for t in Q.index]
   old = Q.index < "2022-04-01"
   S22 = Q.loc[Q.fy == 2022] # FY2021-22 quarters (old-base segment)
   # real GDP: re-link old segment so FY22 equals official old-base FY22 x (new/old FY23 ratio)
   k_r = D.NAS["rgdp_new_FY23"] / D.NAS["rgdp_old_FY23"]
   tgt = D.NAS["rgdp_old_FY22"] * k_r * 10 # crore -> millions
   f_r = tgt / S22.rgdp.sum()
   # nominal GDP
   k_n = D.NAS["ngdp_new"][2023] / D.NAS["ngdp_old"][2023]
   f_n = D.NAS["ngdp_old"][2022] * k_n * 10 / S22.ngdp.sum()
   # PFCE (per-capita x population)
```

```
    pf22_old = D.NAS["pfce_pc_old_FY22"] * D.NAS["pop_FY22_lakh"] * 1e5 / 1e7 # crore
    k_c = D.NAS["pfce_pc_new_FY23"] / D.NAS["pfce_pc_old_FY23"]
    f_c = pf22_old * k_c * 10 / S22.rpfce.sum()
    Q.loc[old, "rgdp"] *= f_r; Q.loc[old, "ngdp"] *= f_n; Q.loc[old, "rpfce"] *= f_c
    Q.attrs["splice_factors"] = dict(real=f_r, nominal=f_n, pfce=f_c, k_real=k_r, k_nom=k_n, k_pfce
        =k_c)
    return Q

def seasonal_adjust(logy, seg_break="2022-04-01"):
    """Seasonal adjustment of a log-level series with a base-year break.
    Old-base segment (2004Q2-2022Q1, 72 obs): robust STL (period 4, seasonal window 7), which
    allows slowly evolving seasonality and down-weights the pandemic outliers.
    New-base segment (2022Q2 onwards, ~17 obs): too short for STL, so stable seasonal factors are
    estimated by OLS of the log level on a linear trend and quarter dummies (sum-to-zero)."""
    import statsmodels.api as sm
    from statsmodels.tsa.seasonal import STL
    sa = logy.copy()
    old = logy.index < seg_break
    s_old = logy[old].dropna()
    sa.loc[s_old.index] = s_old - STL(s_old, period=4, seasonal=7, robust=True).fit().seasonal
    s_new = logy[~old].dropna()
    qd = pd.get_dummies(s_new.index.quarter).astype(float).values
    X = np.column_stack([np.arange(len(s_new)), qd])
    b = np.linalg.lstsq(X, s_new.values, rcond=None)[0]
    sf = b[1:] - b[1:].mean()
    sa.loc[s_new.index] = s_new.values - qd @ sf
    return sa

def monthly_cpi():
    rows = []
    for y, vals in D.cpi_yoy.items():
        for m, v in enumerate(vals, start=1):
            rows.append((pd.Timestamp(y, m, 1), v))
    return pd.Series(dict(rows)).sort_index()

def daily_repo(end="2026-12-31"):
    idx = pd.date_range("2013-01-29", end, freq="D")
    s = pd.Series(np.nan, index=idx)
    for d, r in D.repo_changes:
        s[pd.Timestamp(d)] = r
    return s.ffill()

def quarterly_panel():
    Q = splice_quarterly()
    out = pd.DataFrame(index=Q.index)
    out["fy"] = Q["fy"]
    for v in ["rgdp", "ngdp", "rpfce"]:
        out["l"+v] = np.log(Q[v])
        out["sa_l"+v] = seasonal_adjust(out["l"+v])
    out["g_yoy"] = 100 * (out["sa_lrgdp"] - out["sa_lrgdp"].shift(4)) # real GDP growth y/y (%)
    out["g_qoq"] = 100 * (out["sa_lrgdp"] - out["sa_lrgdp"].shift(1))
    out["c_qoq"] = 100 * (out["sa_lrpfce"] - out["sa_lrpfce"].shift(1))
    out["c_yoy"] = 100 * (out["sa_lrpfce"] - out["sa_lrpfce"].shift(4))
    # NSA y/y within-segment growth (for comparison with official prints)
    out["g_yoy_nsa"] = 100 * (out["lrgdp"] - out["lrgdp"].shift(4))
    cpi = monthly_cpi()
    out = out.reindex(pd.date_range("2004-04-01", "2026-07-01", freq="QS-JAN"))
    out["pi"] = cpi.resample("QS").mean().reindex(out.index)
    rep = daily_repo()
    out["repo_avg"] = rep.resample("QS").mean().reindex(out.index)
    out["repo_end"] = rep.resample("QS").last().reindex(out.index)
    out.attrs = Q.attrs
    return out, cpi, rep

if __name__ == "__main__":
    P, cpi, rep = quarterly_panel()
    print("splice factors:", {k: round(v,4) for k,v in P.attrs["splice_factors"].items()})
```

```
Q = splice_quarterly()
fy = Q.groupby("fy")[["rgdp","ngdp","rpfce"]].sum()/10 # crore
fy = fy[(fy.index>=2006)&(fy.index<=2026)]
print((fy.rgdp.pct_change()*100).round(2).tail(8))
print("nominal FY21 check (should be ~", D.NAS['ngdp_old'][2021]*D.NAS['ngdp_new'][2023]/D.NAS[
    'ngdp_old'][2023], "):", fy.ngdp.loc[2021])
print(P[["g_yoy","g_yoy_nsa","pi","repo_avg"]].loc["2019-01-01":].round(2).to_string())
```

analysis.py

```
"""ECO612 -- full empirical analysis (Q1-Q5). Produces figures/, tables/ and results.json."""
import json, warnings
import numpy as np, pandas as pd
import statsmodels.api as sm
from statsmodels.tsa.filters.hp_filter import hpfilter
from statsmodels.tsa.stattools import adfuller
from statsmodels.stats.diagnostic import acorr_breusch_godfrey
from statsmodels.stats.stattools import durbin_watson
import matplotlib; matplotlib.use("Agg")
import matplotlib.pyplot as plt
import data_inputs as D
from build import quarterly_panel, splice_quarterly, fy_of
warnings.filterwarnings("ignore")
plt.rcParams.update({"font.size":9, "axes.grid":True, "grid.alpha":0.3, "figure.dpi":150,
                     "axes.spines.top":False, "axes.spines.right":False})
R = {} # results container
def q(t): return f"{t.year}Q{(t.month-1)//3+1}"

# ---------------------------------------------------------------- helpers
def ols(y, X, hac=None, const=True):
    X = sm.add_constant(X) if const else X
    y = y.reindex(X.index)
    ok = y.notna() & X.notna().all(axis=1)
    m = sm.OLS(y[ok], X[ok])
    return m.fit(cov_type="HAC", cov_kwds={"maxlags":hac}) if hac else m.fit()

def lincomb(res, w):
    """Linear combination w'b with SE and 95% CI."""
    names = res.params.index
    w = pd.Series(w).reindex(names).fillna(0.0)
    est = float(w @ res.params); se = float(np.sqrt(w @ res.cov_params() @ w))
    return dict(est=est, se=se, lo=est-1.96*se, hi=est+1.96*se)

def ratio_delta(res, num, den_lag):
    """b_num/(1-rho) with delta-method SE."""
    b, r = res.params[num], res.params[den_lag]; V = res.cov_params()
    g = np.array([1/(1-r), b/(1-r)**2])
    Vs = V.loc[[num, den_lag],[num, den_lag]].values
    est = b/(1-r); se = float(np.sqrt(g @ Vs @ g))
    return dict(est=est, se=se, lo=est-1.96*se, hi=est+1.96*se)

def star(p): return "^{***}" if p<0.01 else ("^{**}" if p<0.05 else ("^{*}" if p<0.1 else ""))

def reg_table(models, names, labels, fname, stats_extra=None, caption_note=""):
    """Write booktabs regression table: coef with stars, (SE) below."""
    rows = []
    for v, lab in labels:
        line = [lab]; sel = [""]
        for m in models:
            if v in m.params.index:
                line.append(f"${m.params[v]:.3f}{star(m.pvalues[v])}$"); sel.append(f"$({m.bse[v]:.3f
                    })$")
            else:
                line.append(""); sel.append("")
        rows.append(" & ".join(line)+r" \\"); rows.append(" & ".join(sel)+r" \\")
```

```
    rows.append(r"\midrule")
    rows.append(" & ".join(["Observations"]+[str(int(m.nobs)) for m in models])+r" \\")
    rows.append(" & ".join(["$R^2$ (adj.)"]+[f"{m.rsquared_adj:.3f}" for m in models])+r" \\")
    rows.append(" & ".join(["AIC"]+[f"{m.aic:.1f}" for m in models])+r" \\")
    if stats_extra:
        for lab, vals in stats_extra: rows.append(" & ".join([lab]+vals)+r" \\")
    head = " & ".join([""]+names)+r" \\"
    cols = "l"+"c"*len(models)
    tex = "\\begin{tabular}{%s}\n\\toprule\n%s\n\\midrule\n%s\n\\bottomrule\n\\end{tabular}\n" % (
        cols, head, "\n".join(rows))
    open(f"tables/{fname}.tex","w").write(tex)

def df_table(df, fname, fmt="{:.2f}", index_name=""):
    cols = "l"+"r"*df.shape[1]
    head = " & ".join([index_name]+[str(c) for c in df.columns])+r" \\"
    body = []
    for i, r in df.iterrows():
        cells = [str(i)] + [(fmt.format(x) if isinstance(x,(float,np.floating)) and not np.isnan(x)
            else ("--" if isinstance(x,(float,np.floating)) else str(x))) for x in r.values]
        body.append(" & ".join(cells)+r" \\")
    tex = "\\begin{tabular}{%s}\n\\toprule\n%s\n\\midrule\n%s\n\\bottomrule\n\\end{tabular}\n" % (
        cols, head, "\n".join(body))
    open(f"tables/{fname}.tex","w").write(tex)

# ============================================================== DATA
P, cpi_m, repo_d = quarterly_panel()
R["splice"] = {k: round(float(v),4) for k,v in P.attrs["splice_factors"].items()}

# ============================================================== Q1(a)
s = P.loc["2014-01-01":"2026-07-01"]
fig, ax = plt.subplots(3,1, figsize=(7.2,7.4), sharex=True)
ax[0].plot(s.index, s.g_yoy, color="tab:blue", lw=1.6); ax[0].axhline(0,color="k",lw=.6)
ax[0].set_ylabel("Real GDP growth\n(% y/y, quarterly)")
ax[1].plot(cpi_m.loc["2014":].index, cpi_m.loc["2014":], color="tab:red", lw=1.2)
ax[1].axhspan(2,6,color="grey",alpha=.12); ax[1].axhline(4,color="grey",ls="--",lw=.8)
ax[1].set_ylabel("CPI-C inflation\n(% y/y, monthly)")
rp = repo_d.loc["2014-01-01":"2026-10-31"]
ax[2].step(rp.index, rp.values, where="post", color="tab:green", lw=1.6); ax[2].set_ylabel("Policy
    repo\nrate (%)")
for a in ax:
    for d0,d1 in [("2020-03-01","2021-06-30"),("2022-05-01","2023-02-28"),("2025-02-01","2025-12-31
        ")]:
        a.axvspan(pd.Timestamp(d0),pd.Timestamp(d1),color="orange",alpha=.10)
ax[0].axvspan(pd.Timestamp("2025-04-01"), pd.Timestamp("2026-06-30"), facecolor="none", hatch="///
    ", edgecolor="grey", lw=0, alpha=.5)
ax[0].text(pd.Timestamp("2025-01-15"), -22, "provisional /\nearly est.", fontsize=6.5, ha="right")
ax[1].axvline(pd.Timestamp("2026-01-01"),color="k",ls=":",lw=.8); ax[1].text(pd.Timestamp("
    2026-01-20"),7.5,"CPI base\n2024=100",fontsize=7)
fig.tight_layout(); fig.savefig("figs/fig1_indicators.pdf"); plt.close()

# turning-point quantification
tp = {}
tp["covid"] = dict(g_pre=float(P.loc["2019-10-01","g_yoy"]), g_trough=float(P.loc["2020-04-01","
    g_yoy"]),
                g_rebound=float(P.loc["2021-04-01","g_yoy"]), repo_pre=5.15, repo_post=4.00,
                pi_2020q2=float(P.loc["2020-04-01","pi"]))
tp["tight"] = dict(pi_peak=float(cpi_m.loc["2022"].max()), pi_peak_month=str(cpi_m.loc["2022"].
    idxmax().date())[:7],
                repo_pre=4.00, repo_post=6.50, g_2223=float(P.loc["2022-04-01":"2023-01-01","g_yoy"
                    ].mean()),
                pi_q_2022=float(P.loc["2022-04-01":"2022-10-01","pi"].mean()))
tp["ease"] = dict(pi_trough=float(cpi_m.loc["2025"].min()), pi_trough_month=str(cpi_m.loc["2025"].
    idxmin().date())[:7],
               repo_pre=6.50, repo_post=5.25, g_2025=float(P.loc["2025-01-01":"2025-10-01","g_yoy"
                   ].mean()),
               pi_2025=float(cpi_m.loc["2025"].mean()), pi_sep26=float(cpi_m.iloc[-1]), g_latest=
                   float(P.loc["2026-04-01","g_yoy"]))
```

```
R["turning_points"] = tp
# summary statistics by phase
phases = {"Pre-pandemic (2018Q2--2020Q1)":("2018-04-01","2020-01-01"),
        "Pandemic (2020Q2--2021Q1)":("2020-04-01","2021-01-01"),
        "Recovery (2021Q2--2022Q1)":("2021-04-01","2022-01-01"),
        "Tightening (2022Q2--2023Q1)":("2022-04-01","2023-01-01"),
        "Hold (2023Q2--2024Q4)":("2023-04-01","2024-10-01"),
        "2025 easing (2025Q1--2025Q4)":("2025-01-01","2025-10-01"),
        "2026 (2026Q1--2026Q3)":("2026-01-01","2026-07-01")}
tab = pd.DataFrame({k:[P.loc[a:b,"g_yoy"].mean(), P.loc[a:b,"pi"].mean(), P.loc[a:b,"repo_avg"].
    mean()]
                for k,(a,b) in phases.items()}, index=["Growth","CPI infl.","Repo (avg)"]).T
df_table(tab, "t1_phases", index_name="Phase (mean, \\%)")

# ================================================================== Q1(b) output gap
y = (100*P["sa_lrgdp"]).loc["2004-04-01":"2026-04-01"].dropna()
t = np.arange(len(y))
cyc_hp, tr_hp = hpfilter(y, lamb=1600)
# Hamilton (2018): y_{t+8} on const + y_t..y_{t-3}
H = pd.DataFrame({"y8":y})
for j in range(4): H[f"l{j}"] = y.shift(8+j)
hres = sm.OLS(H.y8, sm.add_constant(H[[f"l{j}" for j in range(4)]]), missing="drop").fit()
gap_ham = (H.y8 - hres.predict(sm.add_constant(H[[f"l{j}" for j in range(4)]]))).reindex(y.index)
# Hamilton robustness: estimate coefficients excluding every observation whose dependent variable
    or
# regressors fall in the pandemic quarters 2020Q2-2021Q2, then apply them to the full sample.
covq = (y.index >= "2020-04-01") & (y.index <= "2021-04-01")
cov_s = pd.Series(covq.astype(float), index=y.index)
touch = cov_s.copy()
for j in range(8, 12): touch = touch + cov_s.shift(j).fillna(0)
Hx = H[touch == 0]
hres_x = sm.OLS(Hx.y8, sm.add_constant(Hx[[f"l{j}" for j in range(4)]]), missing="drop").fit()
gap_ham_x = (H.y8 - hres_x.predict(sm.add_constant(H[[f"l{j}" for j in range(4)]]))).reindex(y.
    index)
# deterministic linear trend with COVID level break (2020Q2 onwards)
Dbrk = (y.index >= "2020-04-01").astype(float)
Xtr = sm.add_constant(np.column_stack([t, Dbrk])); tres = sm.OLS(y.values, Xtr).fit()
gap_lt = pd.Series(y.values - tres.predict(Xtr), index=y.index)
# plain linear trend (no break)
Xl = sm.add_constant(t); gap_lin = pd.Series(y.values - sm.OLS(y.values, Xl).fit().predict(Xl),
    index=y.index)
G = pd.DataFrame({"HP (standard)":cyc_hp, "Hamilton":gap_ham, "Hamilton (ex-COVID est.)":gap_ham_x
    , "Trend+break":gap_lt, "Linear trend":gap_lin})
# sensitivity: HP with COVID quarters interpolated before filtering
y_int = y.copy(); y_int.loc["2020-04-01":"2021-04-01"] = np.nan; y_int = y_int.interpolate()
_, tr_int = hpfilter(y_int, lamb=1600); G["HP"] = y - tr_int
_, tr_int400 = hpfilter(y_int, lamb=400); G["HP ($\\lambda$=400)"] = y - tr_int400
# sensitivity: sample start 2011Q2
_,t11 = hpfilter(y_int.loc["2011-04-01":], lamb=1600); G["HP (from 2011)"] = y - t11
# end-point: one-sided (real-time) HP
rt = {}
for end in y.loc["2018-01-01":].index:
   c,_ = hpfilter(y.loc[:end], lamb=1600); rt[end] = c.iloc[-1]
G["HP real-time"] = pd.Series(rt)
# HP with data ending 2025Q4 (drops 2026 observations)
_,t25 = hpfilter(y_int.loc[:"2025-10-01"], lamb=1600); G["HP (end 2025Q4)"] = (y - t25).loc[:"
    2025-10-01"]
P = P.join(G)
fig, ax = plt.subplots(2,1, figsize=(7.2,5.8))
ax[0].plot(y.index, y, color="k", lw=1.2, label="log real GDP (SA, x100)")
ax[0].plot(y.index, tr_int, color="tab:blue", label="HP trend ($\\lambda$=1600, pandemic quarters
    interpolated)")
ax[0].plot(y.index, tr_hp, color="tab:blue", ls=":", lw=1, label="HP trend, standard")
ax[0].plot(y.index, tres.predict(Xtr), color="tab:orange", ls="--", label="Linear trend + 2020Q2
    break")
ax[0].legend(fontsize=7); ax[0].set_ylabel("log points")
zz = G.loc["2014-01-01":]
```

```
for c, st in [("HP","-"),("Hamilton","-"),("Trend+break","--"),("HP real-time",":")]:
   ax[1].plot(zz.index, zz[c], ls=st, lw=1.3, label=c)
ax[1].axhline(0,color="k",lw=.6); ax[1].set_ylim(-14,12); ax[1].legend(fontsize=7, ncol=4); ax[1].
    set_ylabel("Output gap (% of potential)")
fig.tight_layout(); fig.savefig("figs/fig2_outputgap.pdf"); plt.close()
ep = {"2019Q4 (pre-COVID)":["2019-10-01"], "2020Q2 (trough)":["2020-04-01"], "FY2021-22 avg":["
    2021-04-01","2022-01-01"],
     "FY2022-23 avg":["2022-04-01","2023-01-01"], "2025 avg":["2025-01-01","2025-10-01"], "2026Q2
         (latest)":["2026-04-01"]}
gcols = ["HP","HP (standard)","Hamilton","Hamilton (ex-COVID est.)","Trend+break","Linear trend","
    HP ($\\lambda$=400)","HP (from 2011)","HP (end 2025Q4)","HP real-time"]
gt = pd.DataFrame({k:(G.loc[v[0]:v[-1], gcols].mean()) for k,v in ep.items()}).T
df_table(gt, "t2_gaps", index_name="Episode")
R["gap_corr"] = G[["HP","Hamilton","Trend+break"]].loc["2011-04-01":].corr().round(2).to_dict()
R["gaps_episode"] = gt.round(2).to_dict()
R["ham_coefs"] = hres.params.round(3).to_dict(); R["ham_coefs_exCOVID"] = hres_x.params.round(3).
    to_dict()
R["ham_sum_coefs"] = dict(full=float(hres.params.iloc[1:].sum()), exCOVID=float(hres_x.params.iloc
    [1:].sum()), n_ex=int(hres_x.nobs))
R["ham_x_adf"] = float(adfuller(G["Hamilton (ex-COVID est.)"].loc["2011":].dropna(), autolag="AIC"
    )[1])
R["gap_corr_x"] = float(G[["HP","Hamilton (ex-COVID est.)"]].loc["2011-04-01":].corr().iloc[0,1])
R["trend_break"] = dict(slope_annual=float(4*tres.params[1]), level_break=float(tres.params[2]))

# ================================================================== Q1(c) inflation equation
E = P.loc["2014-01-01":"2026-04-01", ["pi","HP","Hamilton","Trend+break"]].copy()
E["pi_l1"] = E.pi.shift(1)
for g in ["HP","Hamilton","Trend+break"]: E[g+"_l1"] = E[g].shift(1)
E = E.dropna()
def infl(gap, data=E, lagged=True):
   X = data[["pi_l1", gap] + ([gap+"_l1"] if lagged else [])]
   return ols(data.pi, X, hac=4)
m1 = infl("HP"); m0 = infl("HP", lagged=False); m2 = infl("Hamilton"); m3 = infl("Trend+break")
Ex = E.drop(E.loc["2020-04-01":"2021-04-01"].index); m4 = infl("HP", Ex)
R["infl"] = {}
for nm, m, g in [("HP",m1,"HP"),("Ham",m2,"Hamilton"),("TB",m3,"Trend+break"),("HPexCOVID",m4,"HP"
    )]:
   wt = m.wald_test(f"{g} = 0, {g}_l1 = 0", use_f=True, scalar=True)
   comb = lincomb(m, {g:1, g+"_l1":1})
   lr = (comb["est"])/(1-m.params["pi_l1"])
   # delta method for LR
   V = m.cov_params().loc[[g,g+"_l1","pi_l1"],[g,g+"_l1","pi_l1"]].values; r = m.params["pi_l1"]
   grad = np.array([1/(1-r), 1/(1-r), comb["est"]/(1-r)**2]); lrse = float(np.sqrt(grad@V@grad))
   bg = acorr_breusch_godfrey(sm.OLS(m.model.endog, m.model.exog).fit(), nlags=4)
   R["infl"][nm] = dict(F=float(wt.statistic), pF=float(wt.pvalue), comb=comb, lr=lr, lr_se=lrse,
                  bg_p=float(bg[1]), rho=float(r), n=int(m.nobs))
# does lagged slack add explanatory power? (t-test on lag & model comparison)
R["infl"]["lag_add"] = dict(p_lag=float(m1.pvalues["HP_l1"]), aic0=float(m0.aic), aic1=float(m1.
    aic),
                     r2a0=float(m0.rsquared_adj), r2a1=float(m1.rsquared_adj))
# stationarity
R["adf"] = {nm: dict(stat=float(adfuller(sr.dropna(), autolag="AIC")[0]), p=float(adfuller(sr.
    dropna(), autolag="AIC")[1]))
         for nm, sr in [("pi", P.pi.loc["2014":]), ("gap_HP", P.HP.loc["2011":]), ("gap_Ham", P.
             Hamilton.loc["2011":]),
                    ("repo", P.repo_avg.loc["2014":"2026-04-01"]), ("d_repo", P.repo_avg.diff().
                        loc["2014":"2026-04-01"])]}
reg_table([m0,m1,m2,m3,m4], ["(1) HP, no lag","(2) HP","(3) Hamilton","(4) Trend+brk","(5) HP ex-
    COVID"],
       [("const","Constant"),("pi_l1","$\\pi_{t-1}$"),("HP","$\\tilde y_t$"),("HP_l1","$\\tilde
           y_{t-1}$"),
        ("Hamilton","$\\tilde y_t$ (Hamilton)"),("Hamilton_l1","$\\tilde y_{t-1}$ (Hamilton)"),
        ("Trend+break","$\\tilde y_t$ (trend)"),("Trend+break_l1","$\\tilde y_{t-1}$ (trend)")],
       "t3_inflation",
       stats_extra=[("Joint F: gap terms = 0 (p)", ["--"]+[f"{R['infl'][k]['pF']:.3f}" for k in [
           "HP","Ham","TB","HPexCOVID"]]),
```

```
                ("$\\beta_0+\\beta_1$", [f"{m0.params['HP']:.3f}"]+[f"{R['infl'][k]['comb']['est
                    ']:.3f}" for k in ["HP","Ham","TB","HPexCOVID"]]),
                ("Breusch--Godfrey(4) p", [f"{acorr_breusch_godfrey(sm.OLS(m0.model.endog,m0.
                    model.exog).fit(),nlags=4)[1]:.3f}"]+[f"{R['infl'][k]['bg_p']:.3f}" for k in
                     ["HP","Ham","TB","HPexCOVID"]])])

# ================================================================== Q2(a) Taylor rule
TR = P.loc["2014-01-01":"2026-07-01", ["pi","repo_avg","repo_end","HP","Hamilton","Trend+break"]].
    copy()
for c in ["HP","Hamilton","Trend+break"]: TR[c] = TR[c].ffill() # 2026Q3 gap carried forward (GDP
    not yet released)
def taylor(rstar, tp, ty, gap, pistar=4.0): return rstar + TR.pi + tp*(TR.pi-pistar) + ty*TR[gap]
TR["A"] = taylor(1.5,0.5,0.5,"HP"); TR["B"] = taylor(1.0,1.0,0.5,"Hamilton"); TR["C"] = taylor
    (1.9,0.5,1.0,"Trend+break")
for c in "ABC": TR["gap"+c] = TR.repo_avg - TR[c]
fig, ax = plt.subplots(figsize=(7.2,3.6))
ax.plot(TR.index, TR.repo_avg, color="k", lw=2, label="Actual repo (quarter avg)")
ax.plot(TR.index, TR.A, label="Rule A: $r^*$=1.5, $\\theta_\\pi$=0.5, $\\theta_y$=0.5, HP gap")
ax.plot(TR.index, TR.B, label="Rule B: $r^*$=1.0, $\\theta_\\pi$=1.0, $\\theta_y$=0.5, Hamilton
    gap", ls="--")
ax.plot(TR.index, TR.C, label="Rule C: $r^*$=1.9, $\\theta_\\pi$=0.5, $\\theta_y$=1.0, trend gap",
     ls=":")
for d0,d1 in [("2022-04-01","2023-03-31"),("2025-01-01","2025-12-31")]: ax.axvspan(pd.Timestamp(d0
    ),pd.Timestamp(d1),color="orange",alpha=.12)
ax.set_ylim(-8,16); ax.set_ylabel("%"); ax.legend(fontsize=6.5, loc="upper left")
fig.tight_layout(); fig.savefig("figs/fig3_taylor.pdf"); plt.close()
epi = {"Pre-tightening 2021Q4--2022Q1":("2021-10-01","2022-01-01"), "Tightening 2022Q2--2023Q1":("
    2022-04-01","2023-01-01"),
     "Hold 2023Q2--2024Q4":("2023-04-01","2024-10-01"), "Easing 2025Q1--2025Q4":("2025-01-01","
         2025-10-01"),
     "2026Q1--2026Q2":("2026-01-01","2026-04-01"), "2026Q3$^\\dagger$":("2026-07-01","2026-07-01")
         }
tt = pd.DataFrame({k:[TR.loc[a:b,"repo_avg"].mean(), TR.loc[a:b,"pi"].mean(), TR.loc[a:b,"HP"].
    mean(),
                TR.loc[a:b,"A"].mean(), TR.loc[a:b,"gapA"].mean(), TR.loc[a:b,"B"].mean(), TR.loc
                    [a:b,"gapB"].mean(),
                TR.loc[a:b,"C"].mean(), TR.loc[a:b,"gapC"].mean()] for k,(a,b) in epi.items()},
             index=["Repo","$\\pi$","Gap(HP)","$i^{A}$","$i-i^{A}$","$i^{B}$","$i-i^{B}$","$i^{C}
                 $","$i-i^{C}$"]).T
df_table(tt, "t4_taylor", index_name="Episode (means, \\%)")
R["taylor"] = tt.round(2).to_dict()
# one-at-a-time sensitivity: start from Rule A and change a single element
sens = {"Rule A (baseline)":(1.5,0.5,0.5,"HP"), "$r^*=1.0$":(1.0,0.5,0.5,"HP"), "$r^*=1.9$"
    :(1.9,0.5,0.5,"HP"),
     "$\\theta_\\pi=1.0$":(1.5,1.0,0.5,"HP"), "$\\theta_y=1.0$":(1.5,0.5,1.0,"HP"),
     "Hamilton gap":(1.5,0.5,0.5,"Hamilton"), "Trend+break gap":(1.5,0.5,0.5,"Trend+break")}
sw = {"Tightening 2022Q2--2023Q1":("2022-04-01","2023-01-01"), "Easing 2025Q1--2025Q4":("
    2025-01-01","2025-10-01")}
st = {}
for lab,(r_,tp_,ty_,g_) in sens.items():
   ir = taylor(r_,tp_,ty_,g_); row = []
   for (a,b) in sw.values(): row += [ir.loc[a:b].mean(), (TR.repo_avg - ir).loc[a:b].mean()]
   st[lab] = row
st = pd.DataFrame(st, index=["$i^{TR}$ tight.","$i-i^{TR}$ tight.","$i^{TR}$ easing","$i-i^{TR}$
    easing"]).T
df_table(st, "t4b_taylor_sens", index_name="Change from Rule A")
R["taylor_sens"] = st.round(2).to_dict()
R["taylor_BminusA"] = (TR.B - TR.A).groupby([("tight" if "2022-04-01"<=str(d.date())<="2023-01-01"
     else "hold" if "2023-04-01"<=str(d.date())<="2024-10-01" else "ease" if "2025-01-01"<=str(d.
    date())<="2025-10-01" else "x26" if str(d.date())>="2026-01-01" else "other") for d in TR.
    index]).mean().round(2).to_dict()
# change in rule-implied rate vs change in actual over episodes
def chg(col, a, b): return float(TR.loc[b, col] - TR.loc[a, col])
R["taylor_changes"] = {"tight": {c: chg(c,"2022-01-01","2023-01-01") for c in ["repo_avg","A","B",
    "C"]},
                "ease": {c: chg(c,"2024-10-01","2025-10-01") for c in ["repo_avg","A","B","C"]}}
```

```
# ================================================================== Q2(b,c) reaction functions
F = P.loc["2014-01-01":"2026-04-01", ["repo_avg","pi","HP","Hamilton"]].copy()
F["i_l1"] = F.repo_avg.shift(1); F["pi_l1"] = F.pi.shift(1); F["HP_l1"] = F.HP.shift(1); F["pi_l2"
    ]=F.pi.shift(2); F["HP_l2"]=F.HP.shift(2)
F = F.dropna()
rb = ols(F.repo_avg, F[["i_l1","pi","HP"]], hac=4)
rbH = ols(F.repo_avg, F[["i_l1","pi","Hamilton"]].assign(), hac=4)
rc1 = ols(F.repo_avg, F[["i_l1","pi","HP","pi_l1","HP_l1"]], hac=4)
rc2 = ols(F.repo_avg, F[["i_l1","pi_l1","HP_l1"]], hac=4)
Fx = F.drop(F.loc["2020-01-01":"2021-04-01"].index); rbx = ols(Fx.repo_avg, Fx[["i_l1","pi","HP"
    ]], hac=4)
R["rf"] = {}
for nm, m in [("base",rb),("ham",rbH),("lags",rc1),("lagonly",rc2),("exCOVID",rbx)]:
   pv = "pi" if "pi" in m.params.index else "pi_l1"; gv = "HP" if "HP" in m.params.index else ("
       Hamilton" if "Hamilton" in m.params.index else "HP_l1")
   d = dict(rho=float(m.params["i_l1"]), n=int(m.nobs), aic=float(m.aic), bic=float(m.bic), r2a=
       float(m.rsquared_adj),
         bg_p=float(acorr_breusch_godfrey(sm.OLS(m.model.endog,m.model.exog).fit(), nlags=4)[1]))
   if nm != "lags":
      d["LRpi"] = ratio_delta(m, pv, "i_l1"); d["LRy"] = ratio_delta(m, gv, "i_l1")
   R["rf"][nm] = d
# lag model: long-run responses on summed coefficients
r_ = rc1.params["i_l1"]
def ratio_sum(res, nums, den):
   b = sum(res.params[n] for n in nums); r = res.params[den]; V = res.cov_params()
   g = np.array([1/(1-r)]*len(nums) + [b/(1-r)**2]); idx = list(nums)+[den]
   est = b/(1-r); se = float(np.sqrt(g @ V.loc[idx,idx].values @ g))
   return dict(est=est, se=se, lo=est-1.96*se, hi=est+1.96*se)
R["rf"]["lags"]["LRpi"] = ratio_sum(rc1, ["pi","pi_l1"], "i_l1"); R["rf"]["lags"]["LRy"] =
    ratio_sum(rc1, ["HP","HP_l1"], "i_l1")
R["rf"]["lags"]["LRpi_sum"] = R["rf"]["lags"]["LRpi"]["est"]; R["rf"]["lags"]["LRy_sum"] = R["rf"
    ]["lags"]["LRy"]["est"]
# Taylor-principle t-statistics (H0: long-run inflation response = 1) for every column
R["rf_tp_t"] = {k: (R["rf"][k]["LRpi"]["est"]-1)/R["rf"][k]["LRpi"]["se"] for k in ["base","ham","
    lags","lagonly","exCOVID"]}
# Was the 2025 easing in line with the estimated rule? Dynamic simulation of the baseline reaction
     function
# from the 2024Q4 actual repo rate, feeding in actual inflation and the HP gap for 2025Q1-2026Q2.
sim = {}; i_prev = float(P.loc["2024-10-01","repo_avg"])
for d in pd.date_range("2025-01-01","2026-04-01",freq="QS"):
   i_prev = float(rb.params["const"] + rb.params["i_l1"]*i_prev + rb.params["pi"]*P.loc[d,"pi"] +
       rb.params["HP"]*P.loc[d,"HP"])
   sim[q(d)] = dict(sim=i_prev, actual=float(P.loc[d,"repo_avg"]))
R["rf_sim2025"] = sim
# one-step-ahead (static) residuals in 2025
R["rf_resid2025"] = {q(d): float(v) for d,v in rb.resid.loc["2025-01-01":"2025-10-01"].items()}
R["rf"]["lags"]["F_lags"] = float(rc1.wald_test("pi_l1 = 0, HP_l1 = 0", use_f=True, scalar=True).
    pvalue)
# Wald test Taylor principle: LR pi response = 1 <=> beta_pi + rho - 1 = 0
tpw = lincomb(rb, {"pi":1,"i_l1":1}); R["rf"]["taylor_principle"] = dict(est=tpw["est"]-1, se=tpw[
    "se"], t=(tpw["est"]-1)/tpw["se"])
reg_table([rb,rbH,rc1,rc2,rbx], ["(1) Baseline","(2) Hamilton","(3) +lags","(4) lags only","(5) ex
    -COVID"],
       [("const","Constant"),("i_l1","$i_{t-1}$"),("pi","$\\pi_t$"),("pi_l1","$\\pi_{t-1}$"),("HP
           ","$\\tilde y_t$ (HP)"),
        ("HP_l1","$\\tilde y_{t-1}$ (HP)"),("Hamilton","$\\tilde y_t$ (Hamilton)")], "t5_reaction"
            ,
       stats_extra=[("LR $\\pi$ response", [f"{R['rf'][k]['LRpi']['est']:.2f}" if 'LRpi' in R['rf
           ][k] else f"{R['rf'][k]['LRpi_sum']:.2f}" for k in ["base","ham","lags","lagonly","
           exCOVID"]]),
                ("\\quad (s.e.)", [f"({R['rf'][k]['LRpi']['se']:.2f})" for k in ["base","ham","
                    lags","lagonly","exCOVID"]]),
                ("LR gap response", [f"{R['rf'][k]['LRy']['est']:.2f}" if 'LRy' in R['rf'][k]
                    else f"{R['rf'][k]['LRy_sum']:.2f}" for k in ["base","ham","lags","lagonly",
                    "exCOVID"]]),
                ("\\quad (s.e.)", [f"({R['rf'][k]['LRy']['se']:.2f})" for k in ["base","ham","
                    lags","lagonly","exCOVID"]]),
```

```
                ("BIC", [f"{R['rf'][k]['bic']:.1f}" for k in ["base","ham","lags","lagonly","
                    exCOVID"]]),
                ("Breusch--Godfrey(4) p", [f"{R['rf'][k]['bg_p']:.3f}" for k in ["base","ham","
                    lags","lagonly","exCOVID"]])])
# lead-lag: correlation of policy change with lagged/lead inflation and gap
dI = F.repo_avg.diff()
R["leadlag"] = {f"k={k}": dict(pi=float(dI.corr(P.pi.shift(k).reindex(dI.index))), gap=float(dI.
    corr(P.HP.shift(k).reindex(dI.index))))
           for k in [-2,-1,0,1,2,3]}
from statsmodels.tsa.stattools import grangercausalitytests
gc = pd.DataFrame({"di":dI, "dpi":P.pi.diff().reindex(dI.index)}).dropna()
R["granger_pi_to_i"] = float(grangercausalitytests(gc[["di","dpi"]], maxlag=2)[2][0]["ssr_ftest"
    ][1])
R["granger_i_to_pi"] = float(grangercausalitytests(gc[["dpi","di"]], maxlag=2)[2][0]["ssr_ftest"
    ][1])

# ================================================================== Q3 transmission
A = D.rates_A.copy()
def pt(a,b,vars_=("walr_fresh","walr_out","mclr_1y","wadtdr_out","call")):
   di = A.loc[b,"repo"]-A.loc[a,"repo"]
   return {v: dict(chg_bp=100*(A.loc[b,v]-A.loc[a,v]), ratio=(A.loc[b,v]-A.loc[a,v])/di) for v in
       vars_} | {"repo_bp":100*di}
W = {"T1: Mar-22$\\to$Mar-23":pt(2022,2023), "T2: Mar-22$\\to$Mar-24":pt(2022,2024),
    "E1: Mar-25$\\to$Mar-26":pt(2025,2026), "E2: Mar-24$\\to$May-26":pt(2024,2027)}
rows = {}
for k,v in W.items():
   rows[k] = [v["repo_bp"]] + sum([[v[x]["chg_bp"], v[x]["ratio"]] for x in ["walr_fresh","
       walr_out","mclr_1y","wadtdr_out"]], [])
pt_tab = pd.DataFrame(rows, index=["$\\Delta$repo","$\\Delta$WALR-F","PT","$\\Delta$WALR-O","PT ",
    "$\\Delta$MCLR","PT ","$\\Delta$WADTDR-O","PT "]).T
# monthly-release window: Feb-25 -> Jul/Aug-26 (repo 6.50 -> 5.25, -125 bp)
M = D.rates_M.copy(); M.index = pd.PeriodIndex(M.index, freq="M"); M = M.sort_index()
mw = {"walr_fresh":("2025-02","2026-08"), "walr_out":("2025-04","2026-08"), "wadtdr_fresh":("
    2025-02","2026-07"), "wadtdr_out":("2025-02","2026-08")}
# policy change over each window = repo at end of month b minus repo at start of month a
def drepo(a, b): return float(repo_d.loc[:pd.Period(b).end_time].iloc[-1] - repo_d.loc[:pd.Period(
    a).start_time].iloc[-1])
R["pt_monthly"] = {v: dict(a=a,b=b, di=drepo(a,b), chg_bp=100*(M.loc[b,v]-M.loc[a,v]), ratio=(M.
    loc[b,v]-M.loc[a,v])/drepo(a,b)) for v,(a,b) in mw.items()}
pt_tab.loc["E3: Feb-25$\\to$Aug-26$^\\ddagger$"] = [100*R["pt_monthly"]["walr_fresh"]["di"], R["
    pt_monthly"]["walr_fresh"]["chg_bp"], R["pt_monthly"]["walr_fresh"]["ratio"],
    R["pt_monthly"]["walr_out"]["chg_bp"], R["pt_monthly"]["walr_out"]["ratio"], np.nan, np.nan,
    R["pt_monthly"]["wadtdr_out"]["chg_bp"], R["pt_monthly"]["wadtdr_out"]["ratio"]]
df_table(pt_tab, "t6_passthrough", fmt="{:.2f}", index_name="Window")
R["pt"] = {k:{kk:(vv if not isinstance(vv,dict) else {a:round(b,3) for a,b in vv.items()}) for kk,
    vv in v.items()} for k,v in W.items()}
# distributed lag (annual end-March changes)
dA = A.diff(); dA["di_l1"] = dA.repo.shift(1); dA["di_neg"] = dA.repo.where(dA.repo<0, 0.0); dA["
    di_pos"] = dA.repo.where(dA.repo>0,0.0)
dA["di_neg_l1"] = dA.di_neg.shift(1); dA["di_pos_l1"] = dA.di_pos.shift(1)
dl = {}
for v in ["walr_fresh","walr_out","wadtdr_out","mclr_1y"]:
   d = dA[[v,"repo","di_l1","di_pos","di_neg","di_pos_l1","di_neg_l1"]].loc[:2026].dropna()
   m = ols(d[v], d[["repo","di_l1"]])
   cum = lincomb(m, {"repo":1,"di_l1":1})
   ma = ols(d[v], d[["di_pos","di_pos_l1","di_neg","di_neg_l1"]])
   cpos = lincomb(ma, {"di_pos":1,"di_pos_l1":1}); cneg = lincomb(ma, {"di_neg":1,"di_neg_l1":1})
   wt = ma.wald_test("di_pos + di_pos_l1 - di_neg - di_neg_l1 = 0", use_f=True, scalar=True)
   dl[v] = dict(m=m, cum=cum, cpos=cpos, cneg=cneg, asym_p=float(wt.pvalue), n=int(m.nobs),
            dw=float(durbin_watson(m.resid)), ma=ma)
R["dl"] = {v: dict(b0=float(x["m"].params["repo"]), se0=float(x["m"].bse["repo"]), b1=float(x["m"
    ].params["di_l1"]),
              se1=float(x["m"].bse["di_l1"]), cum=x["cum"], cpos=x["cpos"], cneg=x["cneg"], asym_p
                  =x["asym_p"], n=x["n"], dw=x["dw"],
              r2=float(x["m"].rsquared)) for v,x in dl.items()}
reg_table([dl[v]["m"] for v in ["walr_fresh","walr_out","mclr_1y","wadtdr_out"]], ["WALR fresh","
    WALR outst.","1-yr MCLR","WADTDR outst."],
```

```
        [("const","Constant"),("repo","$\\Delta i_t$ ($\\beta_0$)"),("di_l1","$\\Delta i_{t-1}$ ($
            \\beta_1$)")], "t7_dl",
        stats_extra=[("Cumulative $\\beta_0+\\beta_1$", [f"{dl[v]['cum']['est']:.2f}" for v in ["
            walr_fresh","walr_out","mclr_1y","wadtdr_out"]]),
                ("\\quad 95\\% CI", [f"[{dl[v]['cum']['lo']:.2f}, {dl[v]['cum']['hi']:.2f}]" for
                    v in ["walr_fresh","walr_out","mclr_1y","wadtdr_out"]]),
                ("Cum. tightening", [f"{dl[v]['cpos']['est']:.2f}" for v in ["walr_fresh","
                    walr_out","mclr_1y","wadtdr_out"]]),
                ("Cum. easing", [f"{dl[v]['cneg']['est']:.2f}" for v in ["walr_fresh","walr_out",
                    "mclr_1y","wadtdr_out"]]),
                ("Asymmetry test p", [f"{dl[v]['asym_p']:.3f}" for v in ["walr_fresh","walr_out",
                    "mclr_1y","wadtdr_out"]]),
                ("Durbin--Watson", [f"{dl[v]['dw']:.2f}" for v in ["walr_fresh","walr_out","
                    mclr_1y","wadtdr_out"]])])
# figure: rates
fig, ax = plt.subplots(1,2, figsize=(7.4,3.2))
ax[0].plot(A.index, A.repo, "k-o", ms=3, label="Repo"); ax[0].plot(A.index, A.walr_fresh, "-o", ms
    =3, label="WALR fresh")
ax[0].plot(A.index, A.walr_out, "-o", ms=3, label="WALR outstanding"); ax[0].plot(A.index, A.
    wadtdr_out, "-o", ms=3, label="WADTDR outstanding")
ax[0].set_title("End-March, FY2013-14 to May-2026", fontsize=8); ax[0].legend(fontsize=6.5); ax
    [0].set_xticks(range(2014,2028,2)); ax[0].set_xticklabels([f"{y-1}-{str(y)[2:]}" for y in
    range(2014,2028,2)], rotation=45, fontsize=7)
mm = M.copy(); mm.index = mm.index.to_timestamp()
rpm = repo_d.resample("MS").mean().loc["2025-01-01":"2026-09-01"]
ax[1].step(rpm.index, rpm.values, where="post", color="k", label="Repo (monthly avg)")
for c, lab in [("walr_fresh","WALR fresh"),("walr_out","WALR outstanding"),("wadtdr_fresh","WADTDR
     fresh"),("wadtdr_out","WADTDR outstanding")]:
   ax[1].plot(mm.index, mm[c], "o-", ms=3, label=lab)
ax[1].set_title("Monthly releases, 2025-26 easing", fontsize=8); ax[1].legend(fontsize=6.5); ax
    [1].tick_params(axis="x", rotation=45, labelsize=7)
fig.tight_layout(); fig.savefig("figs/fig4_rates.pdf"); plt.close()
# Q3(c) credit lead-lag
cr = pd.Series(D.nonfood, dtype=float); g_cr = 100*cr.pct_change()
g_cr.loc[2024] = 100*(D.nonfood_exmerger[2024]/D.nonfood[2023]-1); g_cr.loc[2025] = 100*(D.
    nonfood_exmerger[2025]/D.nonfood_exmerger[2024]-1)
Qs = splice_quarterly(); FYr = Qs.groupby("fy")[["rgdp","ngdp","rpfce"]].sum()
gFY = 100*FYr.rgdp.pct_change()
C = pd.DataFrame({"gcr":g_cr, "dW":A.walr_fresh.diff(), "dWo":A.walr_out.diff(), "g":gFY}).loc
    [2016:2026]
C["dW_l1"] = C.dW.shift(1); C["dW_f1"] = C.dW.shift(-1); C["dWo_l1"]=C.dWo.shift(1)
R["credit_xcorr"] = {f"corr(gcr_t, dWALR_t{k:+d})": float(C.gcr.corr(C.dW.shift(-k))) for k in
    [-1,0,1]}
Cc = C.dropna(subset=["gcr","dW","dW_l1","g"])
cm = ols(Cc.gcr, Cc[["dW","dW_l1","g"]], hac=1); cm2 = ols(Cc.gcr, Cc[["dWo","dWo_l1","g"]].dropna
    (), hac=1)
R["credit_reg"] = dict(params=cm.params.round(3).to_dict(), se=cm.bse.round(3).to_dict(), n=int(cm
    .nobs), r2=float(cm.rsquared),
                params_o=cm2.params.round(3).to_dict(), se_o=cm2.bse.round(3).to_dict())
R["credit_reg"]["bg_p"] = [float(acorr_breusch_godfrey(sm.OLS(m.model.endog,m.model.exog).fit(),
    nlags=1)[1]) for m in (cm, cm2)]
reg_table([cm,cm2], ["WALR fresh","WALR outst."], [("const","Constant"),("dW","$\\Delta$WALR$_t$")
    ,("dW_l1","$\\Delta$WALR$_{t-1}$"),
       ("dWo","$\\Delta$WALR$^{O}_t$"),("dWo_l1","$\\Delta$WALR$^{O}_{t-1}$"),("g","Real GDP
           growth$_t$")], "t8_credit",
       stats_extra=[("Breusch--Godfrey(1) p", [f"{p:.3f}" for p in R["credit_reg"]["bg_p"]]),
                ("Durbin--Watson", [f"{durbin_watson(m.resid):.2f}" for m in (cm, cm2)])])
R["credit_series"] = {int(k): round(float(v),2) for k,v in g_cr.loc[2016:].items()}

# ============================================================================ Q4 fiscal
FI = D.FISC.copy().astype(float)
# official-basis nominal GDP: old base to FY22, new base from FY23; FY27 = BE assumption (10% over
     FY26 PE)
fyN = FYr.ngdp/10 # linked (new-base level) crore
kN = R["splice"]["k_nom"]
ngdp_off = (fyN/kN).loc[2005:2022].to_dict()
```

```
ngdp_off.update(D.NAS["ngdp_new"]); ngdp_off[2027] = D.NAS["ngdp_new"][2026]*(1+D.
    NOMINAL_GROWTH_BE_2027)
ngdp_link = fyN.loc[2005:2026].to_dict(); ngdp_link.update({k:v for k,v in D.NAS["ngdp_new"].items
    ()}); ngdp_link[2027] = ngdp_off[2027]
FI["Y"] = pd.Series(ngdp_off); FI["Ylink"] = pd.Series(ngdp_link)
for c in ["GFD","GPD","RD","REVEX","CAPEX","INT","PIT","DIR","IND"]: FI[c+"_y"] = 100*FI[c]/FI.Y
FI["PB_y"] = -FI.GPD_y
# annual output gap = FY mean of quarterly HP gap (FY27: 2026Q2 only)
gapFY = P["HP"].groupby(P.fy).mean(); gapFY_ham = P["Hamilton"].groupby(P.fy).mean()
FI["gap"] = gapFY; FI["gap_ham"] = gapFY_ham
for eps, tag in [(0.10,""),(0.20,"_e2")]:
   FI["CAPB"+tag] = FI.PB_y - eps*FI.gap
   FI["FI"+tag] = -FI["CAPB"+tag].diff()
FI["CAPB_ham"] = FI.PB_y - 0.10*FI.gap_ham; FI["FI_ham"] = -FI.CAPB_ham.diff()
FI["dgap"] = FI.gap.diff()
ft = FI.loc[2015:2027, ["GFD_y","GPD_y","RD_y","CAPEX_y","REVEX_y","gap","CAPB","FI","FI_e2"]].
    copy()
ft.index = [f"{y-1}-{str(y)[2:]}" + ("$^{RE}$" if y==2026 else ("$^{BE}$" if y==2027 else "")) for
     y in ft.index]
ft.columns = ["GFD","GPD","RD","CapEx","RevEx","Gap","CAPB","FI($\\varepsilon$=.1)","FI($\\
    varepsilon$=.2)"]
df_table(ft, "t9_fiscal", index_name="FY")
R["fiscal"] = FI.loc[2015:2027, ["GFD_y","GPD_y","RD_y","CAPEX_y","REVEX_y","gap","CAPB","FI","
    FI_e2","FI_ham"]].round(2).to_dict()
# cyclicality: corr(FI, gap_{t}) and corr(FI, gap_{t-1}) over FY2016-FY2027
cyc = FI.loc[2016:2027]
R["fiscal_cyc"] = dict(corr_FI_gap=float(cyc.FI.corr(cyc.gap)), corr_FI_gaplag=float(cyc.FI.corr(
    cyc.gap.shift(1))),
                corr_FI_dgap=float(cyc.FI.corr(cyc.dgap)))
fig, ax = plt.subplots(1,2, figsize=(7.4,3.2))
xx = FI.loc[2015:2027]
ax[0].bar(xx.index-0.2, xx.CAPEX_y, 0.4, label="Capital exp.")
ax[0].bar(xx.index+0.2, xx.REVEX_y-xx.INT_y, 0.4, label="Revenue exp. ex-interest")
ax[0].plot(xx.index, xx.GFD_y, "k-o", ms=3, label="Gross fiscal deficit")
ax[0].set_ylabel("% of GDP"); ax[0].legend(fontsize=6.5); ax[0].set_title("Centre: composition",
    fontsize=8)
ax[1].bar(xx.index, xx.FI, color=["tab:red" if v>0 else "tab:blue" for v in xx.FI], label="Fiscal
    impulse (+ = expansionary)")
ax[1].plot(xx.index, xx.gap, "k-o", ms=3, label="Output gap (HP, FY avg)"); ax[1].axhline(0,color=
    "k",lw=.6)
ax[1].legend(fontsize=6.5); ax[1].set_title("Fiscal impulse vs cycle", fontsize=8)
for a in ax: a.set_xticks(range(2015,2028,2)); a.set_xticklabels([f"{y-1}-{str(y)[2:]}" for y in
    range(2015,2028,2)], rotation=45, fontsize=7)
fig.tight_layout(); fig.savefig("figs/fig5_fiscal.pdf"); plt.close()
# Q4(b,c) expenditure-growth regressions
X4 = pd.DataFrame(index=range(2006,2027))
X4["g"] = gFY; X4["g_l1"] = gFY.shift(1)
YL = FI.Ylink
X4["dk"] = 100*(FI.CAPEX - FI.CAPEX.shift(1))/YL.shift(1)
X4["dr"] = 100*((FI.REVEX-FI.INT) - (FI.REVEX-FI.INT).shift(1))/YL.shift(1)
X4["dk_l1"] = X4.dk.shift(1); X4["dr_l1"] = X4.dr.shift(1)
X4["D21"] = (X4.index==2021).astype(float); X4["D22"] = (X4.index==2022).astype(float)
X4 = X4.loc[2007:2026]
f1 = ols(X4.g, X4[["g_l1","dk","dr","D21","D22"]], hac=1)
f0 = ols(X4.g, X4[["g_l1","dk","dr"]], hac=1)
f2 = ols(X4.g, X4[["g_l1","dk","dr","dk_l1","dr_l1","D21","D22"]].dropna(), hac=1)
f3 = ols(X4.g, X4[["g_l1","dk_l1","dr_l1","D21","D22"]].dropna(), hac=1)
R["exp"] = dict(
  b=dict(bK=float(f1.params.dk), seK=float(f1.bse.dk), bR=float(f1.params.dr), seR=float(f1.bse.dr
      ),
       pdiff=float(f1.wald_test("dk = dr", use_f=True, scalar=True).pvalue), n=int(f1.nobs)),
  c=dict(cumK=lincomb(f2, {"dk":1,"dk_l1":1}), cumR=lincomb(f2, {"dr":1,"dr_l1":1}),
       pdiffcum=float(f2.wald_test("dk + dk_l1 = dr + dr_l1", use_f=True, scalar=True).pvalue),
       lagK=float(f2.params.dk_l1), lagK_se=float(f2.bse.dk_l1), lagR=float(f2.params.dr_l1),
           lagR_se=float(f2.bse.dr_l1),
       bg_p=float(acorr_breusch_godfrey(sm.OLS(f2.model.endog,f2.model.exog).fit(), nlags=1)[1])),
  sd_dk=float(X4.dk.std()), sd_dr=float(X4.dr.std()))
```

```
reg_table([f0,f1,f2,f3], ["(1)","(2)","(3) + lags","(4) lags only"],
       [("const","Constant"),("g_l1","$g_{t-1}$"),("dk","$\\Delta k_t$"),("dr","$\\Delta r_t$"),(
           "dk_l1","$\\Delta k_{t-1}$"),
        ("dr_l1","$\\Delta r_{t-1}$"),("D21","FY2020-21 dummy"),("D22","FY2021-22 dummy")], "
            t10_expgrowth",
       stats_extra=[("Wald $\\beta_K=\\beta_R$ (p)", [f"{f0.wald_test('dk = dr', use_f=True,
           scalar=True).pvalue:.3f}", f"{R['exp']['b']['pdiff']:.3f}", f"{R['exp']['c']['pdiffcum
           ']:.3f}", "--"]),
             ("Cum. capital ($\\Sigma\\beta_K$)", ["--","--", f"{R['exp']['c']['cumK']['est
                 ']:.2f} ({R['exp']['c']['cumK']['se']:.2f})", "--"]),
             ("Cum. revenue ($\\Sigma\\beta_R$)", ["--","--", f"{R['exp']['c']['cumR']['est
                 ']:.2f} ({R['exp']['c']['cumR']['se']:.2f})", "--"]),
             ("Breusch--Godfrey(1) p", [f"{acorr_breusch_godfrey(sm.OLS(m.model.endog,m.model.
                 exog).fit(), nlags=1)[1]:.3f}" for m in (f0,f1,f2,f3)])])
R["exp_data_2026"] = X4.loc[2023:2026, ["g","dk","dr"]].round(2).to_dict()

# ================================================================== Q5(a) consumption
C5 = P.loc["2005-04-01":"2026-04-01", ["c_qoq","g_qoq","c_yoy","g_yoy","repo_avg"]].copy()
C5["g_l1"] = C5.g_qoq.shift(1); C5["drepo_l1"] = C5.repo_avg.diff().shift(1)
for d in ["2020-04-01","2020-07-01","2021-04-01","2021-07-01"]: C5["D"+d[:7]] = (C5.index==d).
    astype(float)
dums = [c for c in C5.columns if c.startswith("D20")]
C5r = C5.loc["2014-01-01":].dropna(subset=["drepo_l1"])
c1 = ols(C5.c_qoq, C5[["g_qoq","g_l1"]+dums].dropna(), hac=4)
c2 = ols(C5r.c_qoq, C5r[["g_qoq","g_l1","drepo_l1"]+dums], hac=4)
C5["gy_l4"] = C5.g_yoy.shift(4)
c3 = ols(C5.c_yoy, C5[["g_yoy","gy_l4"]].dropna(), hac=4)
Cx = C5.drop(C5.loc["2020-01-01":"2021-10-01"].index); c4 = ols(Cx.c_qoq, Cx[["g_qoq","g_l1"]].
    dropna(), hac=4)
R["cons"] = {nm: dict(b0=float(m.params.iloc[1]), se0=float(m.bse.iloc[1]), b1=float(m.params.iloc
    [2]), se1=float(m.bse.iloc[2]),
                 comb=lincomb(m, {m.params.index[1]:1, m.params.index[2]:1}), n=int(m.nobs))
          for nm, m in [("qoq",c1),("qoq_rate",c2),("yoy",c3),("exCOVID",c4)]}
R["cons"]["bg_p"] = {nm: float(acorr_breusch_godfrey(sm.OLS(m.model.endog,m.model.exog).fit(),
    nlags=4)[1]) for nm,m in [("qoq",c1),("qoq_rate",c2),("yoy",c3),("exCOVID",c4)]}
R["cons"]["sd_ratio"] = float(C5.loc["2011":].drop(C5.loc["2020":"2021"].index).c_qoq.std()/C5.loc
    ["2011":].drop(C5.loc["2020":"2021"].index).g_qoq.std())
reg_table([c1,c2,c4,c3], ["(1) q/q","(2) q/q + rate","(3) q/q ex-COVID","(4) y/y"],
       [("const","Constant"),("g_qoq","$\\Delta Y_t$"),("g_l1","$\\Delta Y_{t-1}$"),("drepo_l1","
           $\\Delta i_{t-1}$"),
        ("g_yoy","$\\Delta_4 Y_t$"),("gy_l4","$\\Delta_4 Y_{t-4}$")], "t11_consumption",
       stats_extra=[("$\\beta_0+\\beta_1$", [f"{R['cons'][k]['comb']['est']:.2f}" for k in ["qoq"
           ,"qoq_rate","exCOVID","yoy"]]),
             ("\\quad 95\\% CI", [f"[{R['cons'][k]['comb']['lo']:.2f}, {R['cons'][k]['comb']['
                 hi']:.2f}]" for k in ["qoq","qoq_rate","exCOVID","yoy"]]),
             ("COVID dummies", ["Yes","Yes","Sample excl.","No"]),
             ("Breusch--Godfrey(4) p", [f"{R['cons']['bg_p'][k]:.3f}" for k in ["qoq","
                 qoq_rate","exCOVID","yoy"]])])
# annual: MPC out of GDP (first differences of real levels, Rs crore) and growth elasticities
A5 = pd.DataFrame({"C":FYr.rpfce/1e6, "Y":FYr.rgdp/1e6}).loc[2005:2026]
A5["dC"] = A5.C.diff(); A5["dY"] = A5.Y.diff(); A5["dY_l1"] = A5.dY.shift(1)
A5["D21"] = (A5.index==2021).astype(float); A5["D22"] = (A5.index==2022).astype(float)
c5 = ols(A5.dC, A5[["dY","dY_l1","D21","D22"]], hac=1)
R["cons"]["mpc"] = dict(b0=float(c5.params.dY), se0=float(c5.bse.dY), b1=float(c5.params.dY_l1),
    se1=float(c5.bse.dY_l1),
                comb=lincomb(c5, {"dY":1,"dY_l1":1}), n=int(c5.nobs), apc=float((A5.C/A5.Y).loc
                    [2016:2026].mean()))
reg_table([c5], ["$\\Delta C_t$ (Rs lakh crore)"], [("const","Constant"),("dY","$\\Delta Y_t$"),("
    dY_l1","$\\Delta Y_{t-1}$"),("D21","FY2020-21"),("D22","FY2021-22")], "t11b_mpc",
       stats_extra=[("$\\beta_0+\\beta_1$", [f"{R['cons']['mpc']['comb']['est']:.2f} ({R['cons
           ']['mpc']['comb']['se']:.2f})"]),
             ("Average $C/Y$, FY16--26", [f"{R['cons']['mpc']['apc']:.2f}"])])
# Q5(b) taxes and consumption (annual)
T5 = pd.DataFrame(index=range(2006,2027))
T5["c"] = 100*FYr.rpfce.pct_change(); T5["g"] = gFY
T5["dpit"] = (100*FI.PIT/FI.Ylink).diff(); T5["dind"] = (100*FI.IND/FI.Ylink).diff(); T5["ddir"] =
     (100*FI.DIR/FI.Ylink).diff()
```

```
T5["dpit_l1"] = T5.dpit.shift(1); T5["c_l1"] = T5.c.shift(1)
T5["D21"] = (T5.index==2021).astype(float); T5["D22"] = (T5.index==2022).astype(float)
T5 = T5.loc[2007:2026]
t1m = ols(T5.c, T5[["dpit","dpit_l1","D21","D22"]], hac=1)
t2m = ols(T5.c, T5[["dpit","dpit_l1","g","D21","D22"]], hac=1)
t3m = ols(T5.c, T5[["dpit","dind","g","D21","D22"]], hac=1)
t4m = ols(T5.c, T5[["ddir","dind","g","D21","D22"]], hac=1)
# narrative dummy for fiscal years with a major household-facing tax cut: FY2019-20 (Rs 5 lakh rebate, interim
# Budget 2019), FY2023-24 (new regime default, Rs 7 lakh rebate), FY2025-26 (Rs 12 lakh threshold + GST 2.0 rate cuts)
T5["TAXCUT"] = T5.index.isin([2020, 2024, 2026]).astype(float)
t5m = ols(T5.c, T5[["dpit","dpit_l1","g","D21","D22","TAXCUT"]], hac=1)
R["tax"] = {nm: {k: dict(b=float(m.params[k]), se=float(m.bse[k]), p=float(m.pvalues[k])) for k in
     m.params.index if k not in ["const","D21","D22"]} | {"n":int(m.nobs)}
         for nm,m in [("m1",t1m),("m2",t2m),("m3",t3m),("m4",t4m),("m5",t5m)]}
R["tax"]["m2"]["cum"] = lincomb(t2m, {"dpit":1,"dpit_l1":1})
R["tax"]["pit_y"] = (100*FI.PIT/FI.Y).loc[2022:2027].round(2).to_dict()
R["tax"]["ind_y"] = (100*FI.IND/FI.Y).loc[2023:2027].round(2).to_dict()
R["tax"]["bg_p"] = [float(acorr_breusch_godfrey(sm.OLS(m.model.endog,m.model.exog).fit(), nlags=1)[1]) for m in (t1m,t2m,t3m,t4m,t5m)]
reg_table([t1m,t2m,t3m,t4m,t5m], ["(1)","(2)","(3)","(4)","(5)"],
        [("const","Constant"),("dpit","$\\Delta(T^{PIT}/Y)_t$"),("dpit_l1","$\\Delta(T^{PIT}/Y)_{t-1}$"),("ddir","$\\Delta(T^{DIR}/Y)_t$"),
         ("dind","$\\Delta(T^{IND}/Y)_t$"),("g","Real GDP growth$_t$"),("TAXCUT","Tax-relief year dummy"),("D21","FY2020-21 dummy"),("D22","FY2021-22 dummy")], "t12_tax",
        stats_extra=[("Breusch--Godfrey(1) p", [f"{p:.3f}" for p in R["tax"]["bg_p"]])])
# ADF unit-root tests for the regression variables of Q1-Q5 (AIC lag selection; constant only)
def adf_row(name, sr, freq, sample):
    # quarterly: lag length by AIC; annual (T<=20): one fixed lag, since AIC selection is unreliable in tiny samples
    sr = sr.dropna()
    st_, p_, lag_, n_ = (adfuller(sr, autolag="AIC") if freq=="Q" else adfuller(sr, maxlag=1, autolag=None))[:4]
    return [name, freq, sample, f"{st_:.2f}", f"{p_:.3f}", str(lag_), str(n_)]
ur = [adf_row("Inflation $\\pi_t$", P.pi.loc["2014":], "Q", "2014Q1--2026Q3"),
     adf_row("Output gap (HP)", P.HP.loc["2011":], "Q", "2011Q1--2026Q2"),
     adf_row("Output gap (Hamilton)", P.Hamilton.loc["2011":], "Q", "2011Q1--2026Q2"),
     adf_row("Output gap (Hamilton, ex-COVID est.)", P["Hamilton (ex-COVID est.)"].loc["2011":], "Q", "2011Q1--2026Q2"),
     adf_row("Repo rate $i_t$", P.repo_avg.loc["2014":"2026-04-01"], "Q", "2014Q1--2026Q2"),
     adf_row("$\\Delta i_t$", P.repo_avg.diff().loc["2014":"2026-04-01"], "Q", "2014Q2--2026Q2"),
     adf_row("$\\Delta$WALR fresh", A.walr_fresh.diff(), "A", "FY16--FY27"),
     adf_row("$\\Delta$WALR outstanding", A.walr_out.diff(), "A", "FY15--FY27"),
     adf_row("$\\Delta$WADTDR outstanding", A.wadtdr_out.diff(), "A", "FY15--FY27"),
     adf_row("Credit growth", g_cr.loc[2016:2026], "A", "FY16--FY26"),
     adf_row("Real GDP growth $g_t$ (annual)", X4.g, "A", "FY07--FY26"),
     adf_row("$\\Delta k_t$", X4.dk, "A", "FY07--FY26"),
     adf_row("$\\Delta r_t$", X4.dr, "A", "FY07--FY26"),
     adf_row("$\\Delta C_t$ (q/q)", C5.c_qoq, "Q", "2005Q3--2026Q2"),
     adf_row("$\\Delta Y_t$ (q/q)", C5.g_qoq, "Q", "2005Q3--2026Q2"),
     adf_row("PFCE growth (annual)", T5.c, "A", "FY07--FY26"),
     adf_row("$\\Delta(T^{PIT}/Y)$", T5.dpit, "A", "FY07--FY26")]
ut = pd.DataFrame([r[1:] for r in ur], index=[r[0] for r in ur], columns=["Freq.","Sample","ADF stat.","p-value","Lags","Obs."])
df_table(ut, "t13_adf", index_name="Series")
R["adf_all"] = {r[0]: float(r[4]) for r in ur}
R["sample_info"] = dict(q_infl=f"{q(E.index[0])}--{q(E.index[-1])}", q_rf=f"{q(F.index[0])}--{q(F.index[-1])}",
                q_cons=f"{q(C5.dropna(subset=['g_l1']).index[0])}--{q(C5.index[-1])}")
# Summary for Q5(c)
R["summary"] = dict(gap_2025_HP=float(P.loc["2025-01-01":"2025-10-01","HP"].mean()), gap_2025_Ham=float(P.loc["2025-01-01":"2025-10-01","Hamilton"].mean()),
            gap_2025_TB=float(P.loc["2025-01-01":"2025-10-01","Trend+break"].mean()),
                gap_latest=float(P.loc["2026-04-01","HP"]))
json.dump(R, open("results.json","w"), indent=1, default=lambda o: float(o) if isinstance(o,(np.floating,np.integer)) else str(o))
```

```
# export analysis datasets for the appendix/replication
P.to_csv("quarterly_dataset.csv"); FI.to_csv("annual_fiscal_dataset.csv"); A.to_csv("
    annual_rates_dataset.csv")
print("done")
```